

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Drawing

Semester: Spring

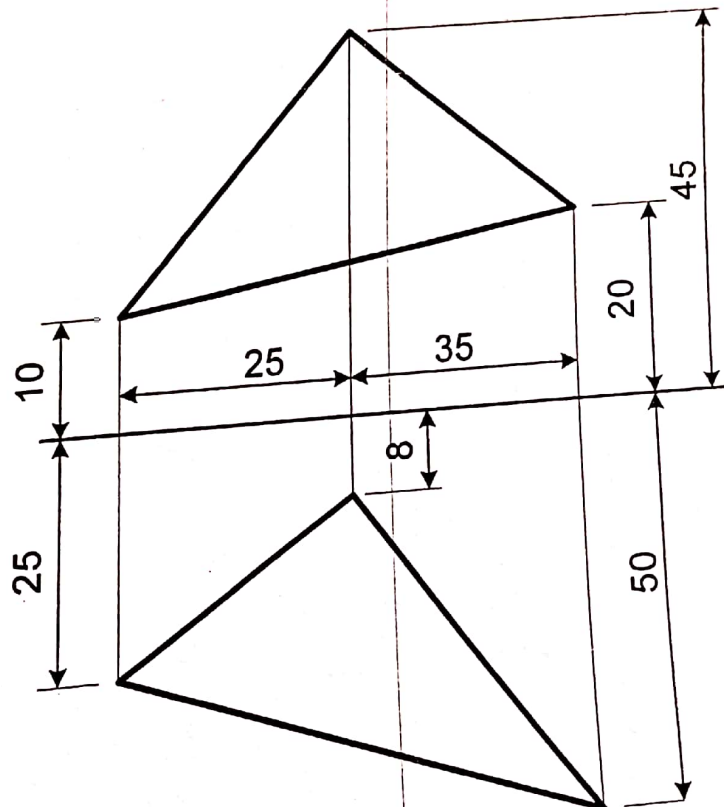
Year : 2012
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. Draw an ellipse by auxiliary or concentric circles method, given the major and minor axes as 80 and 50 mm respectively. 14
2. Draw the true shape of the plane from the projections of the plane as shown. 14

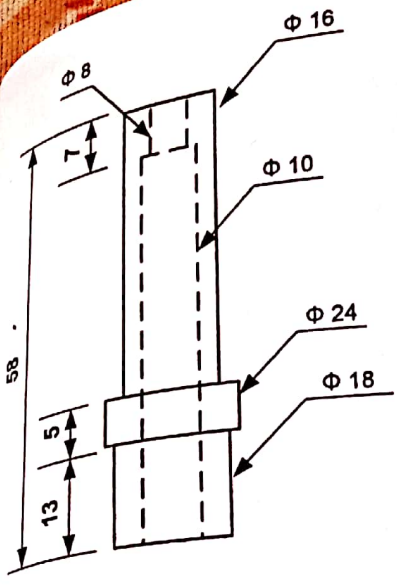


3. Draw complete orthographic views (three views) of the given object (figure 1). 30

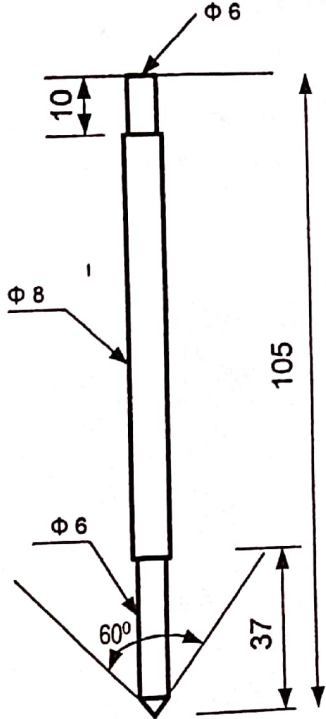
Level
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Course

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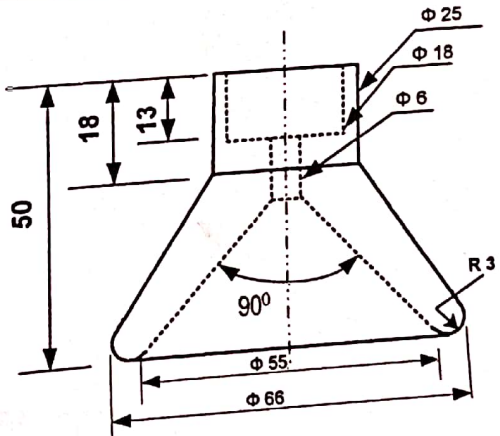
1. D
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2. E
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Guide



Centering Pin



Centering Cone

Figure 4

5.

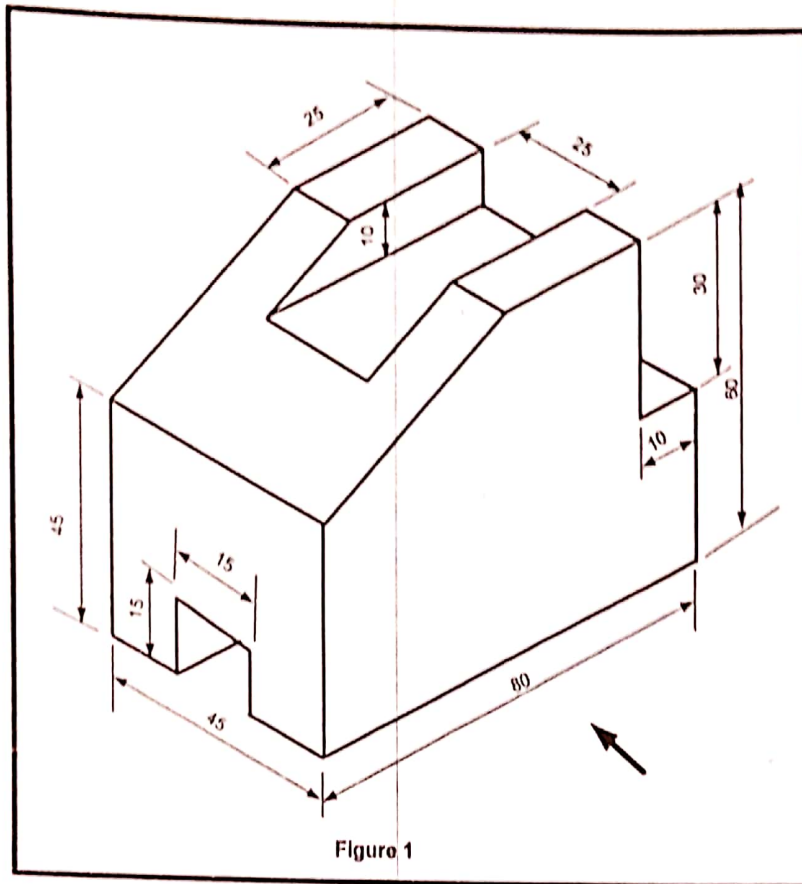


Figure 1

4. Draw the given views in figure 2 with curve of intersection.

18

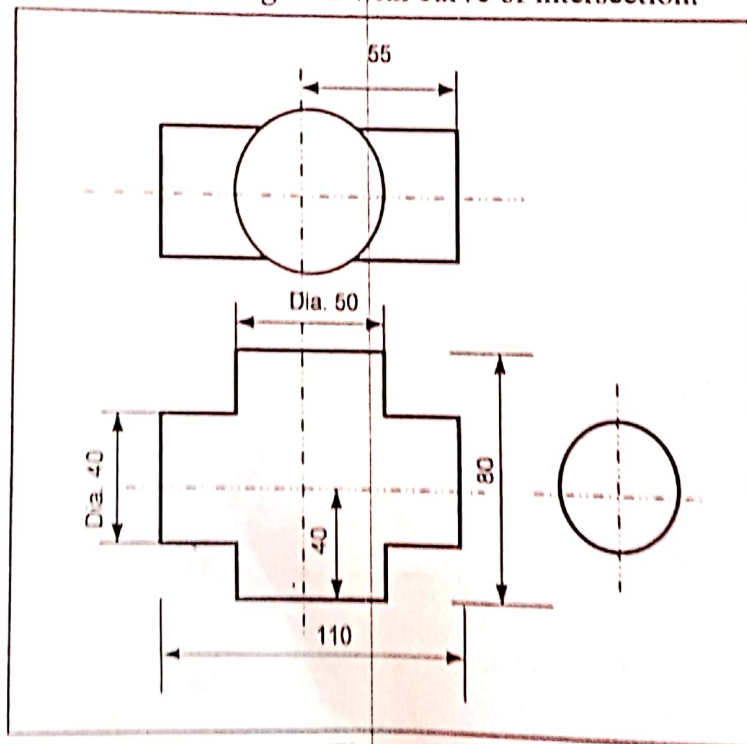
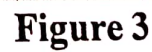


Figure 2

- 18

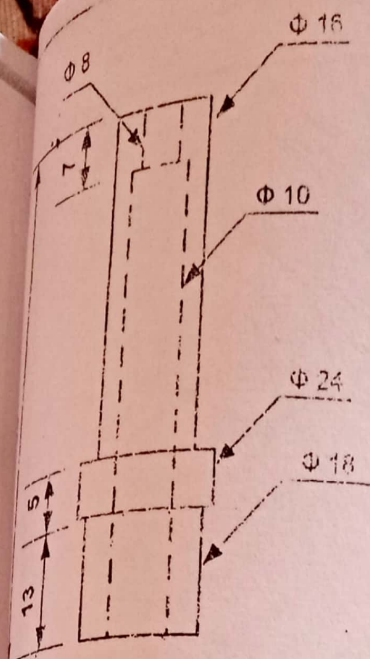


OR

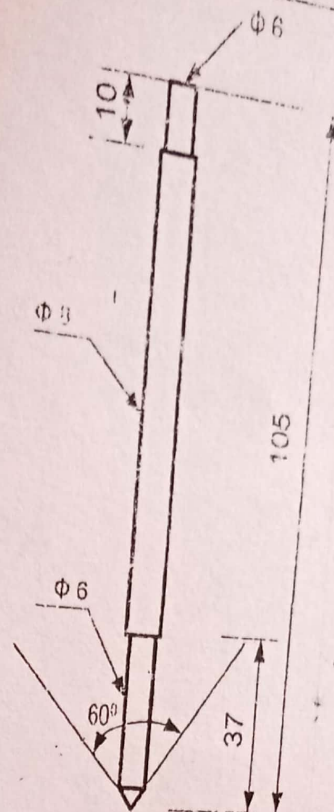
OR

Assemble the given parts of the Centering Cone and draw its full sectional (orthographic) front view. (figure 4)

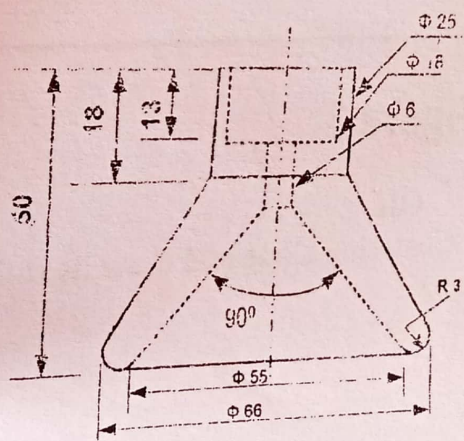
or a small
al view



Guide



Centering Pin



Centering Cone

Figure 4

full

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE Civil
Course: Engineering Drawing

Semester – Fall

Year : 2012
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks. All dimensions are in mm.
Attempt all the questions.

Construct a cycloid of a circle having diameter 40 mm. 14

A regular pentagon **ABCDE** of 25 mm sides has its corner **A** in HP and the side **CD** parallel to **HP**. Draw its projections when its plane is parallel to and 10 mm from the **VP**. 14

Draw complete orthographic views (three views) of the given object (figure 1). 30

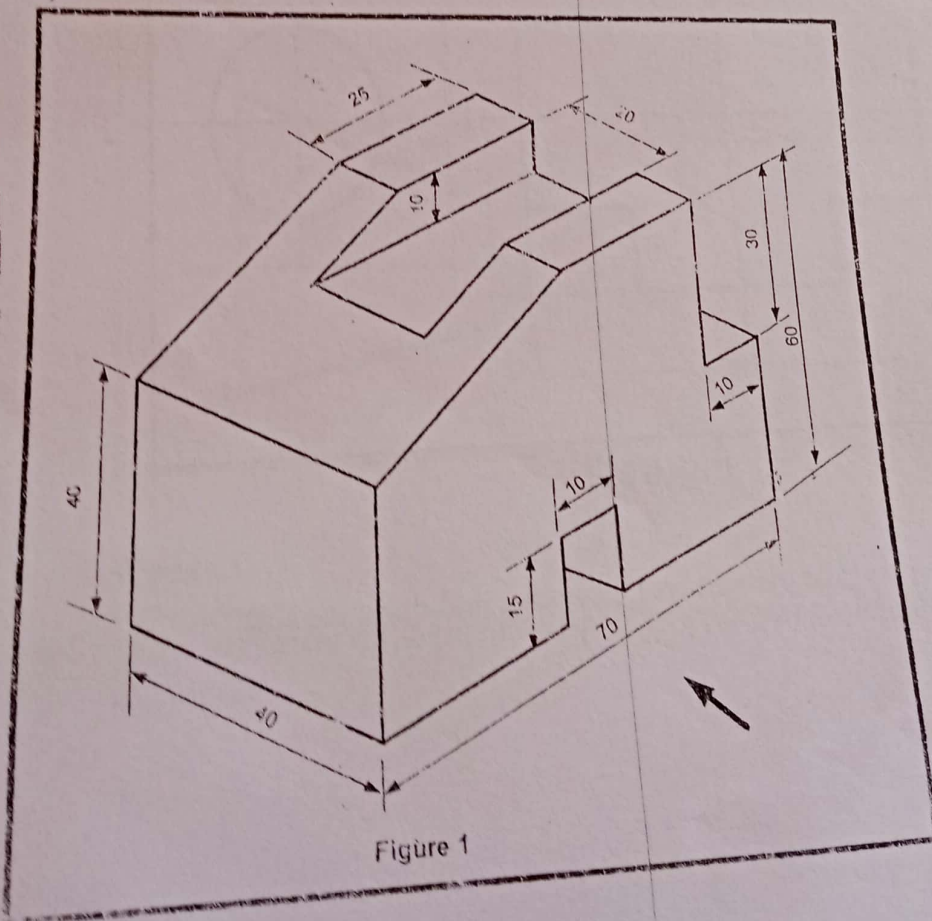


Figure 1

Draw the given views in figure 2 with curve of intersection.

18

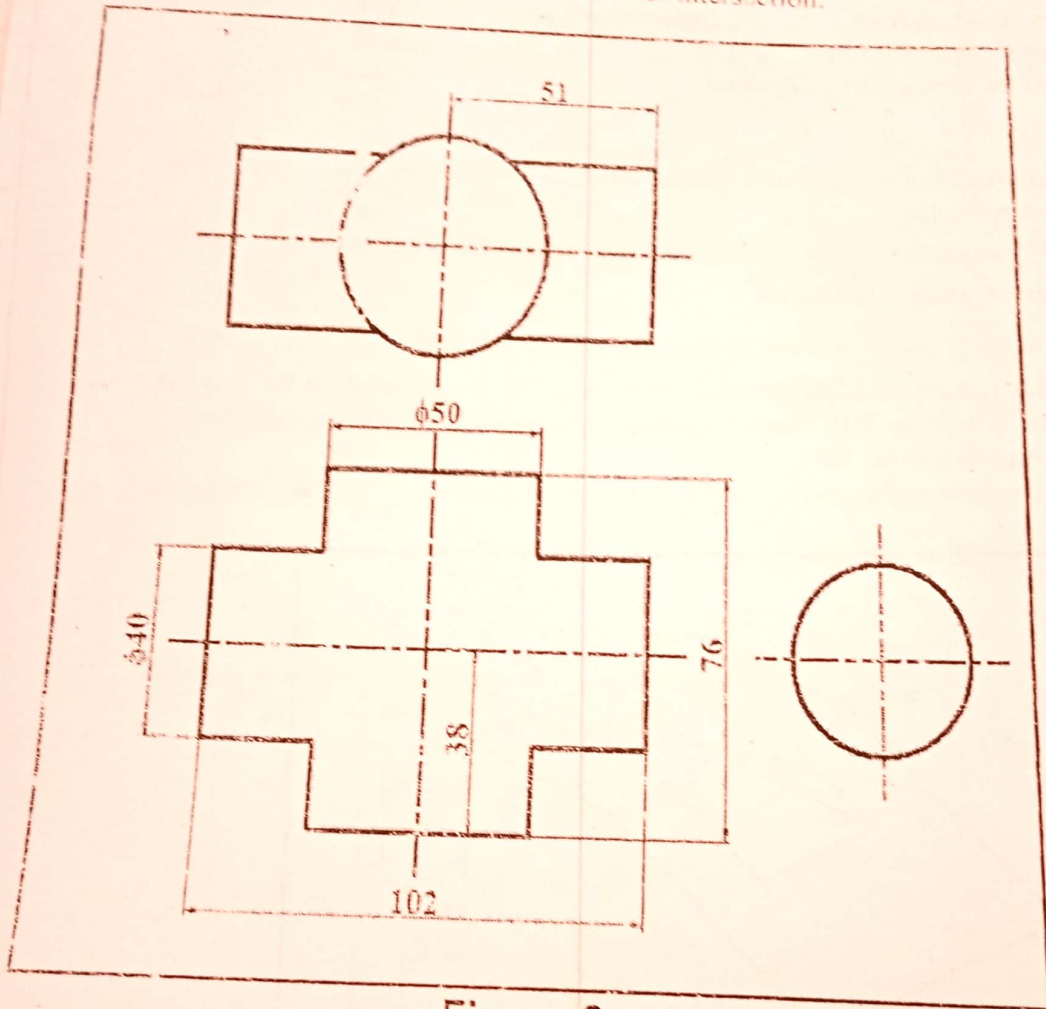


Figure 2

Assemble the given parts of the universal coupling and draw the full sectional (orthographic) front view (figure 3).

24

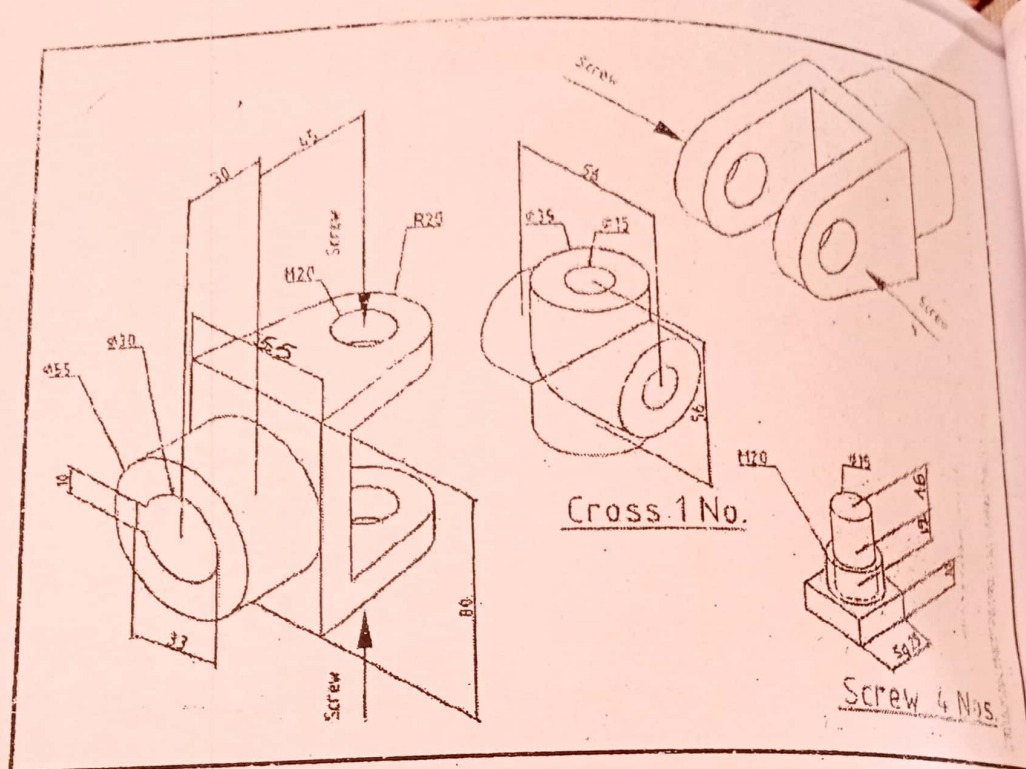


Figure 3

OR

Assembled the given parts of the Centering Cone and draw its full sectional (orthographic) front view. (figure 4)

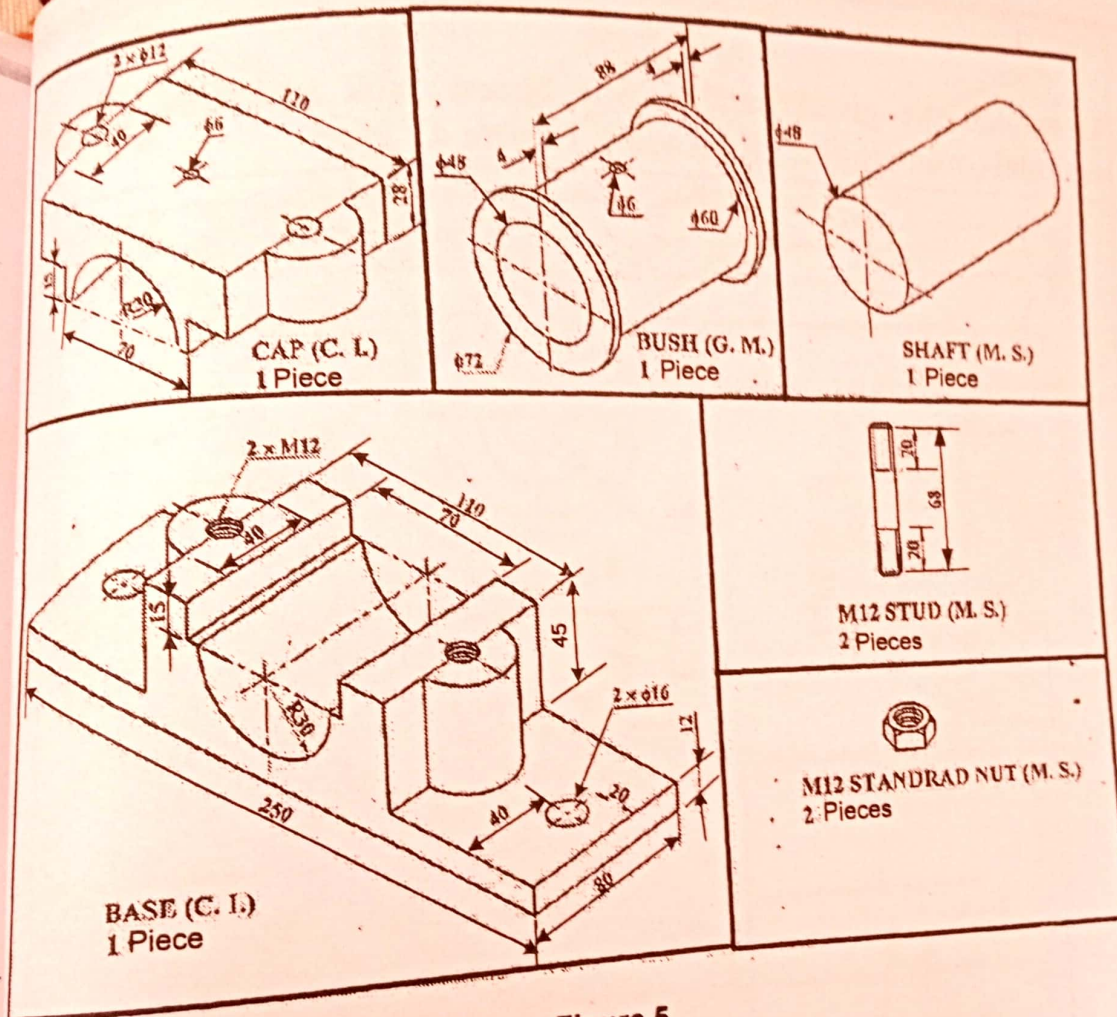


Figure 5

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Drawing

Semester: Spring

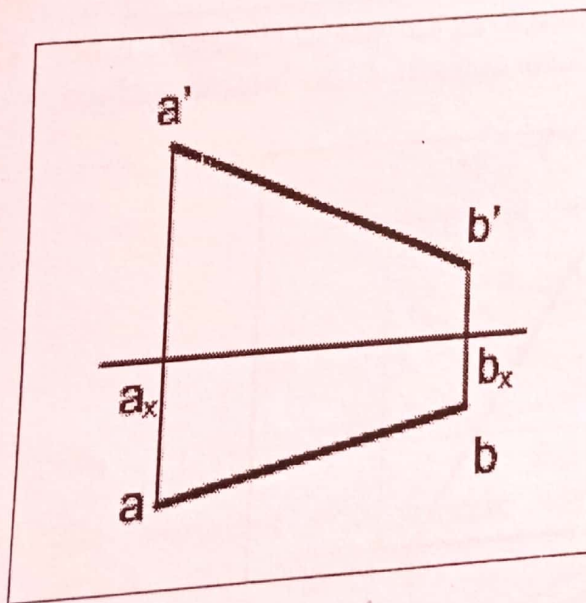
Year : 2012
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. Construct a uncrossed tangent line (open belt type) between two circles having diameter 50 mm and 30 mm. and center to center distance between them is 80 mm. 14
2. Reproduce the given views of the lines and determine the true length. (figure 1) 14

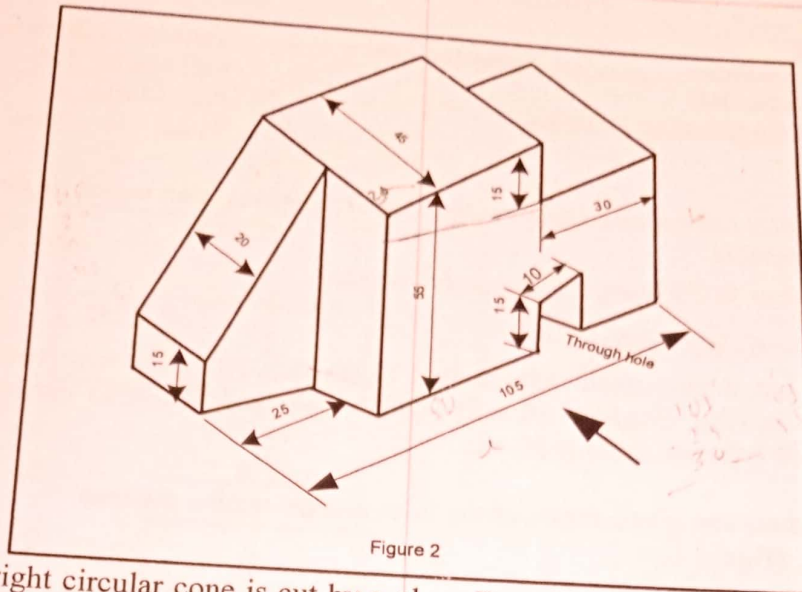


$$\begin{aligned} a'a_x &= 30 \\ aa_x &= 25 \\ a_xb_x &= 45 \end{aligned}$$

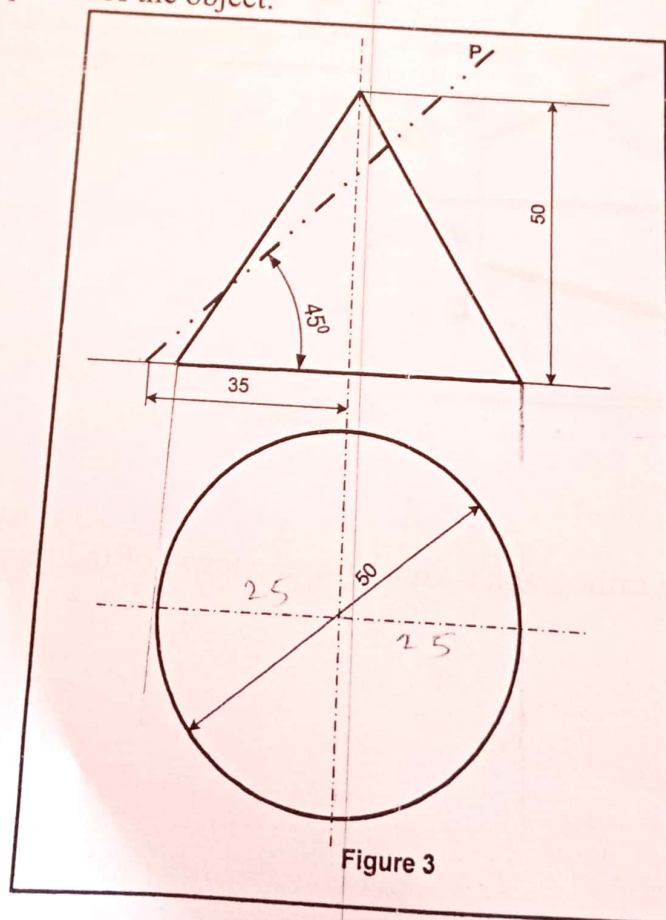
$$\begin{aligned} b'b_x &= 15 \\ bb_x &= 20 \end{aligned}$$

Figure 1

3. Draw complete orthographic views (three views) of the given object (figure 2). 30



4. A right circular cone is cut by a plane **P** as shown in figure 3. Find the true shape of the section and also construct the lateral surface development of the object. 18



5. Assemble the given parts of the Screw Jack and draw its full sectional (orthographic) front view. (figure 4)

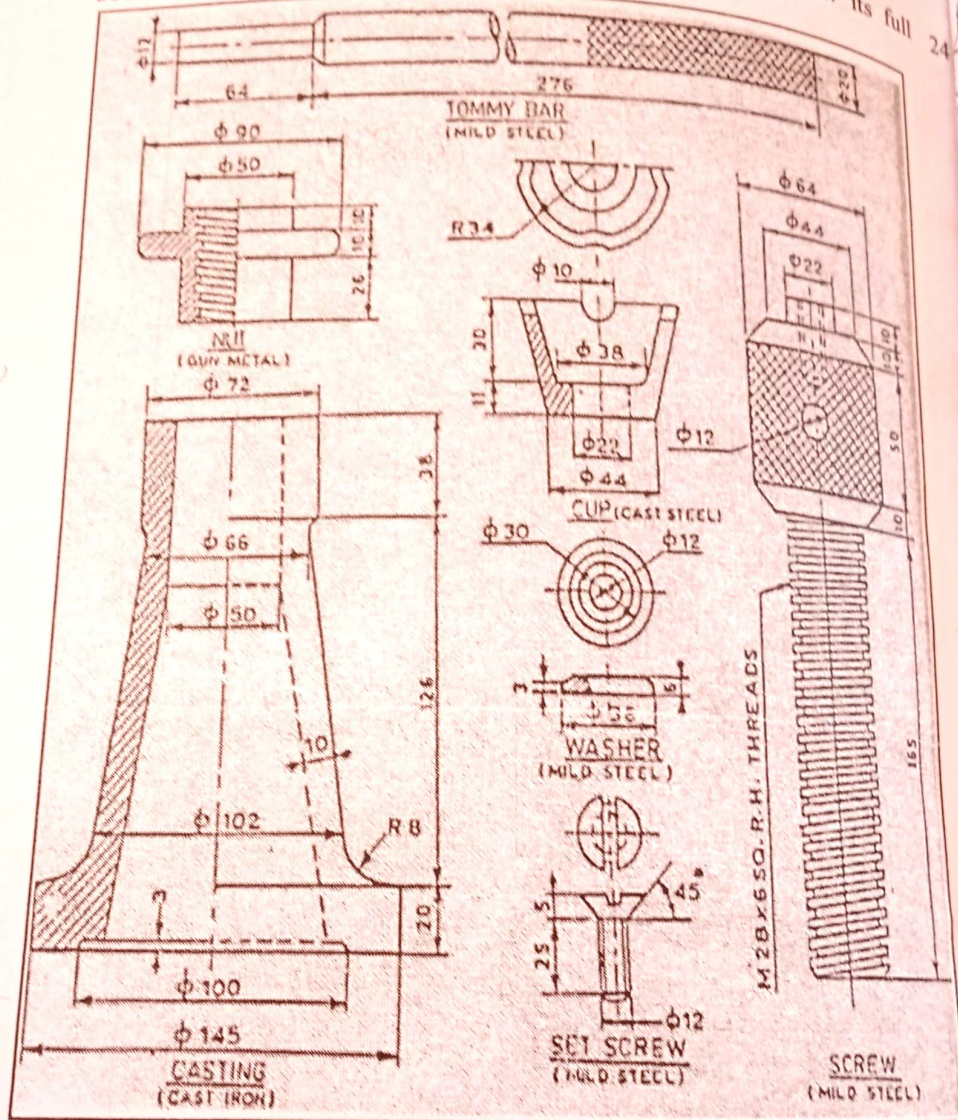


Figure 4

OR

Assemble the given parts of the split bearing and draw its full sectional (orthographic) front view. (figure 5)

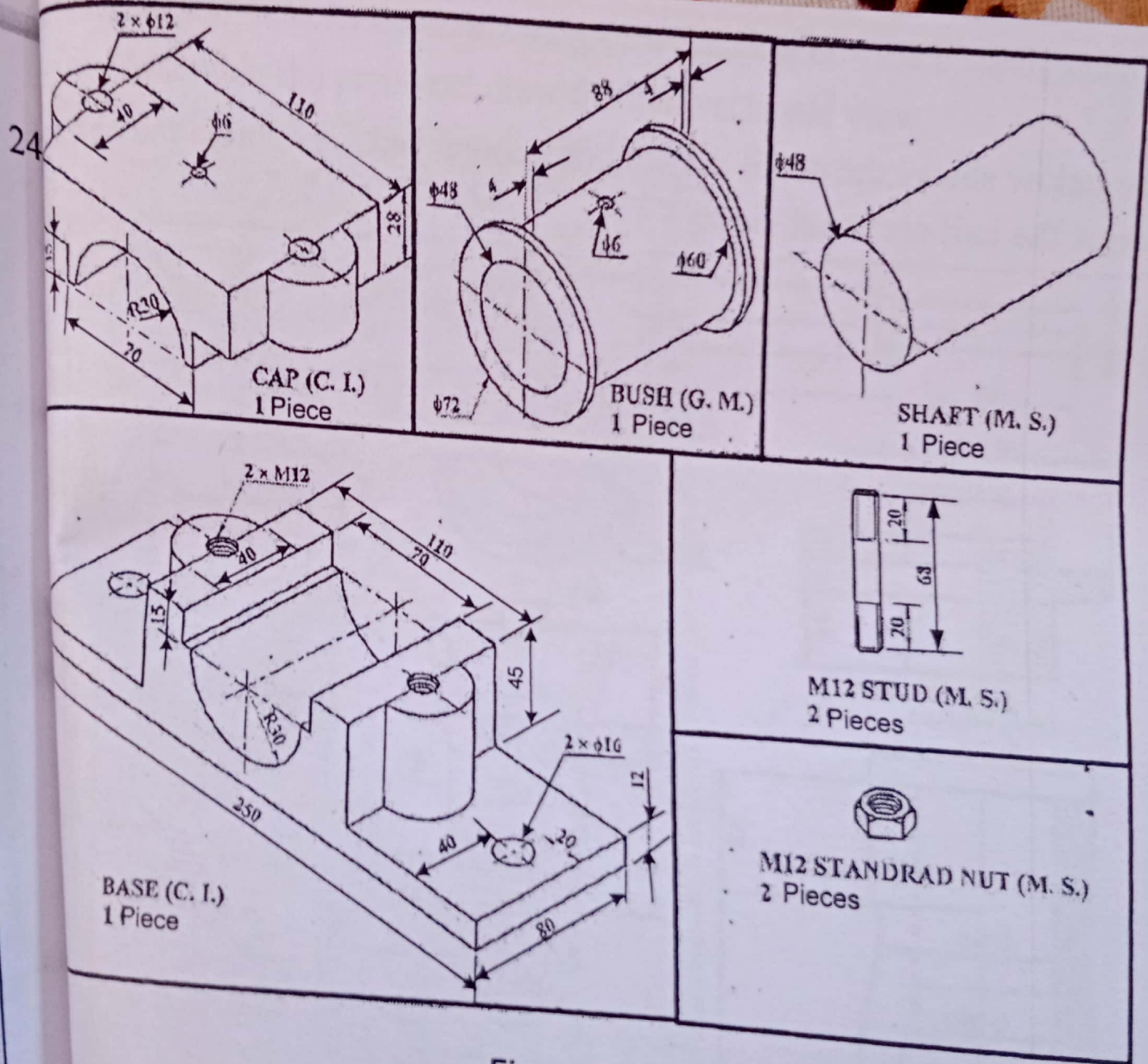


Figure 5

POKHARA UNIVERSITY

Level: Bachelor
 Programme: BE
 Course: Engineering Drawing

Semester: Fall

Year : 2013
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Draw an **ellipse** having major axis 90 mm and minor axis 60 mm by concentric circle method. 15
- b) Draw a **hexagon** having sides of length 40 mm. 5
2. Reproduce the given views of the plane and determine the true shape. (Figure1)..All dimensions are in mm. 8

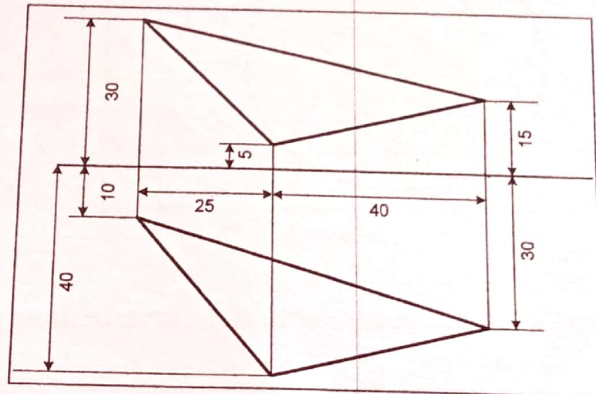


Figure 1

3. Draw the complete orthographic view of the following figure with sectional side view. (figure 2) 30

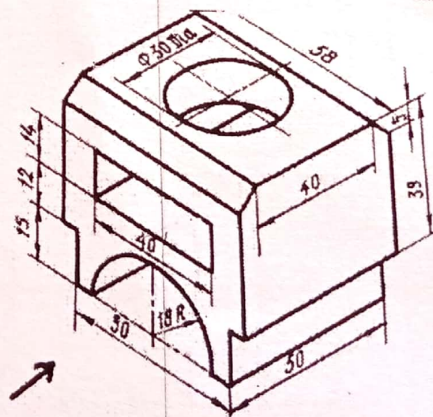


Figure 2

4. A right circular cone is cut by a plane P as shown in figure below. Find the true shape of the section and also construct the lateral surface development of the object. (figure 3) 18

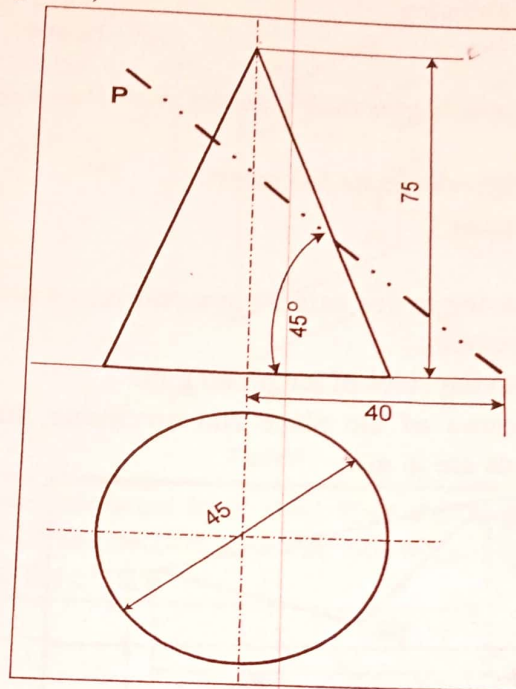


Figure 3

OR

Draw the given views of in figure below with the curve of intersection. (figure 4)

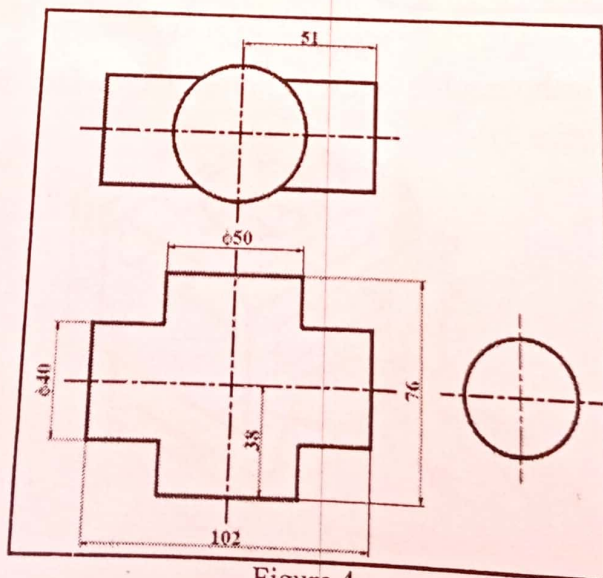
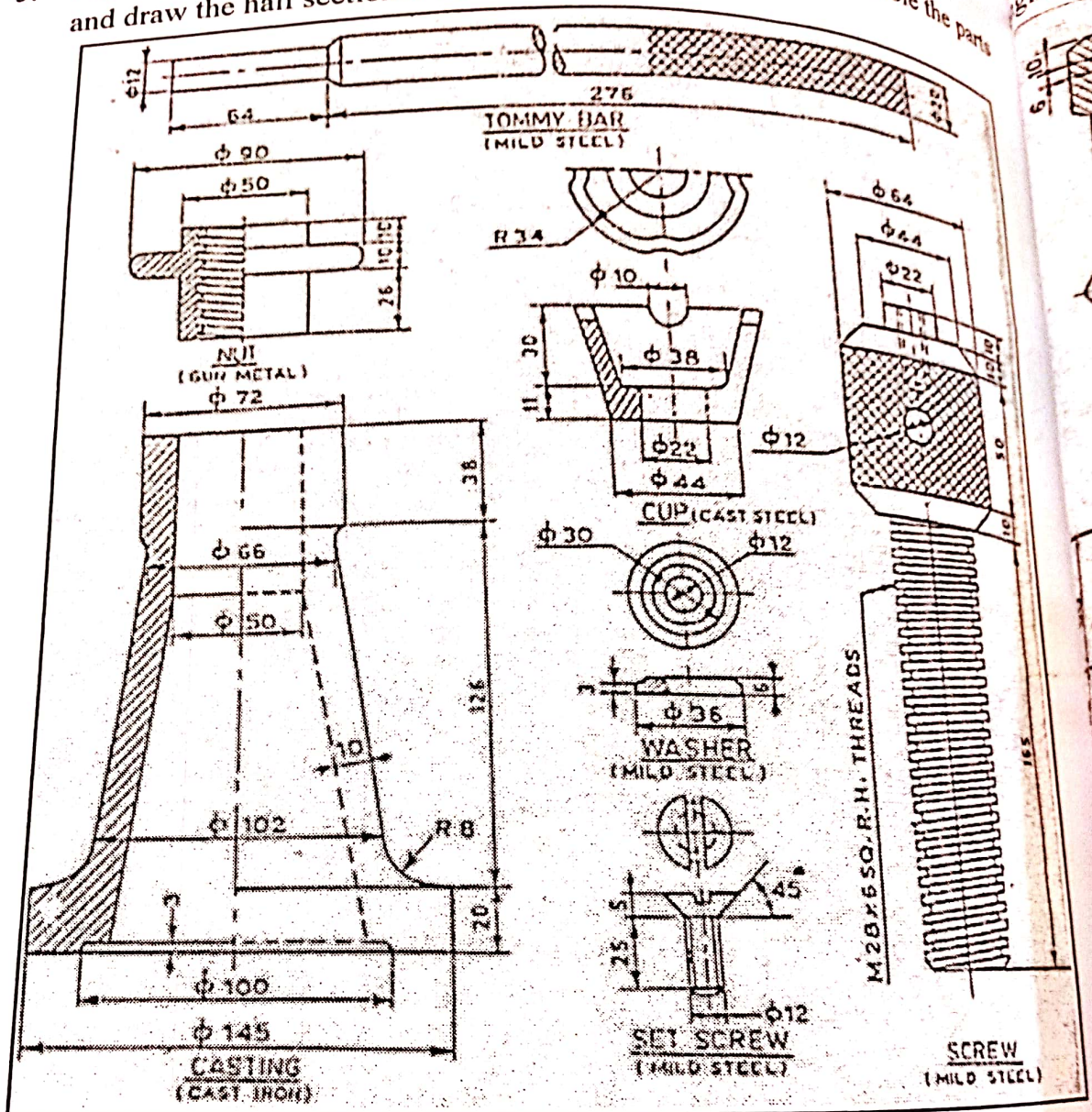


Figure 4

the 18
ent

5. Figure below shows the detail drawing of a screw jack. Assemble the parts and draw the half sectional view.

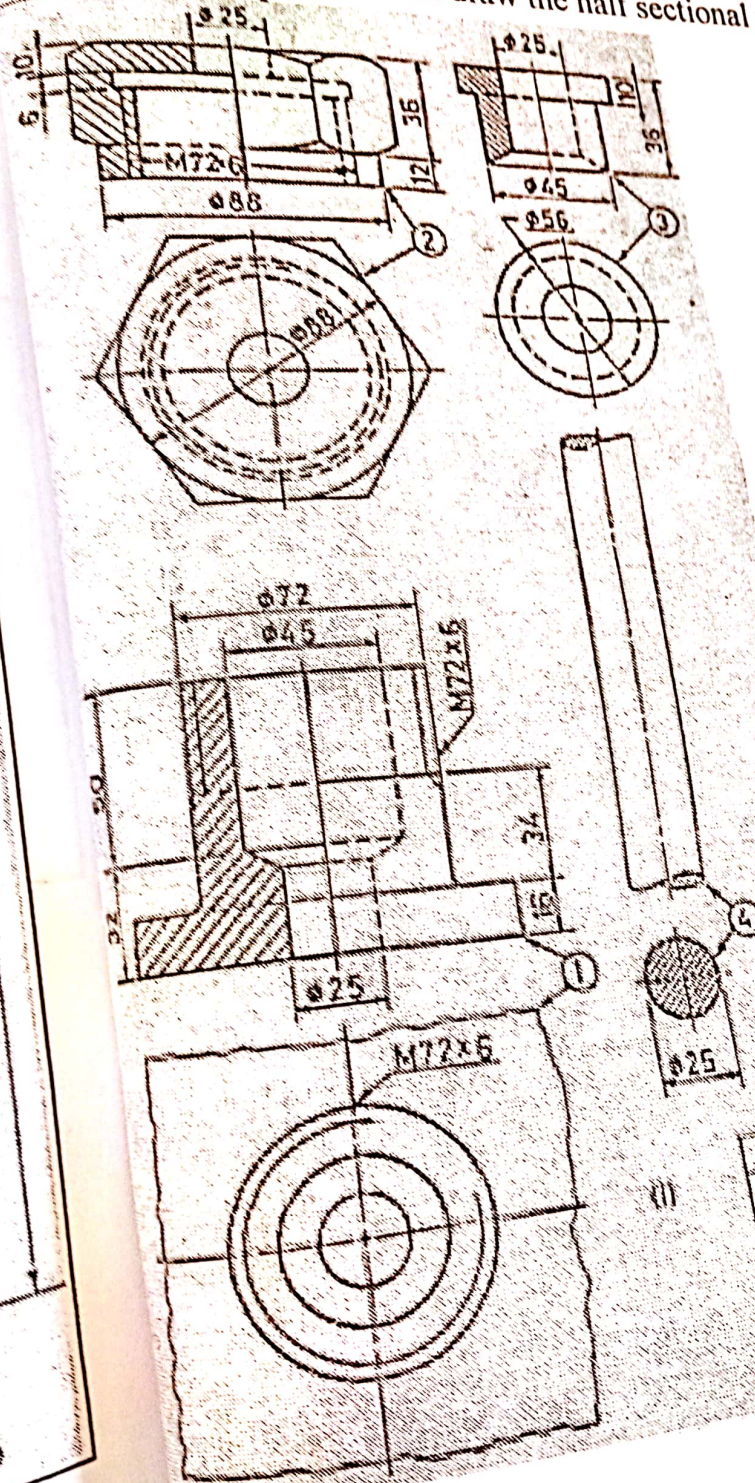
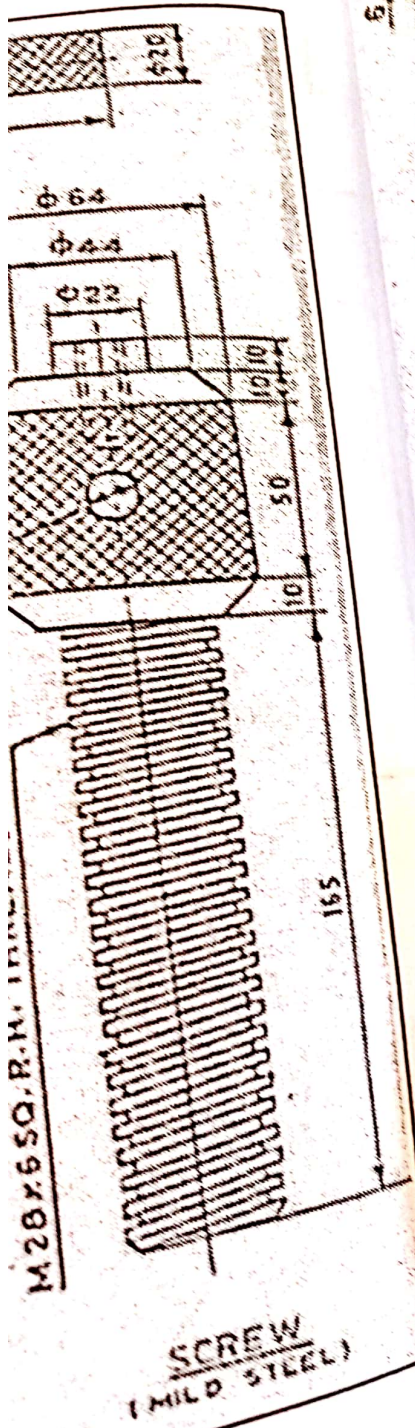


OR

Figure below shows the detail drawing of a split bearing. Assemble the parts and draw the half sectional view.

Figure below shows the detail drawing of a Stuffing box for a small steam engine. Assemble the parts and draw the half sectional view.

Assemble the parts



ALL DIMENSIONS IN MM

SCALE: 1:0

Sl. No.	Name of part	Qty	Material
5	Packing	1	Asbestos
4	Piston rod	1	C-30
3	Gland bush	1	Brass
2	Nut	1	C.I.
1	Cylinder	1	C.I.

POKHARA UNIVERSITY

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The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Draw an **involute** of a circle having radius of 25 mm. 10
b) Draw a **Pentagon** having sides of length 40 mm. 10
2. Reproduce the given views of the lines and determine the true length. 8

$a'a_x = 10\text{mm}$	$b'b_x = 30\text{mm}$
$a_x a = 45\text{mm}$	$b_x b = 20\text{mm}$
$a_x b_x = 80\text{mm}$	

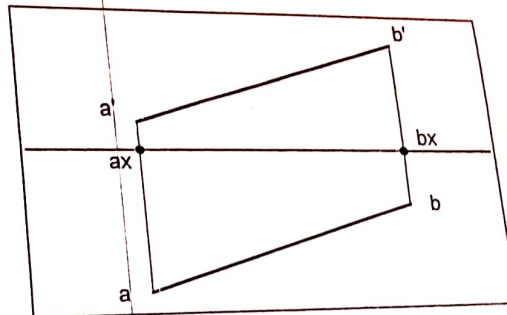


Figure 1

3. Draw the complete orthographic view of the following figure with sectional side view. 30

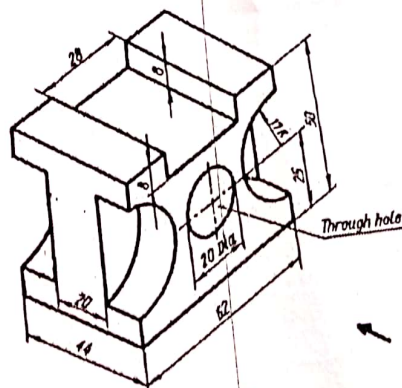


Figure 2

4. A pyramid is cut by a Plane P as shown in the figure below. Find the true 18

shape of the section and construct the lateral surface development of the object.

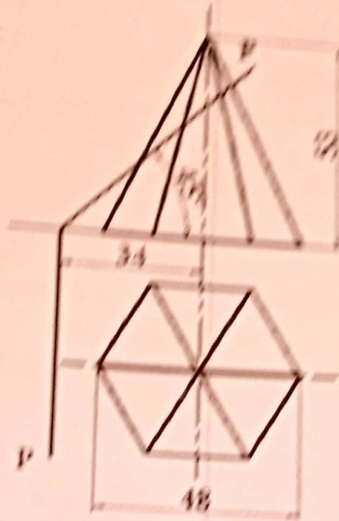
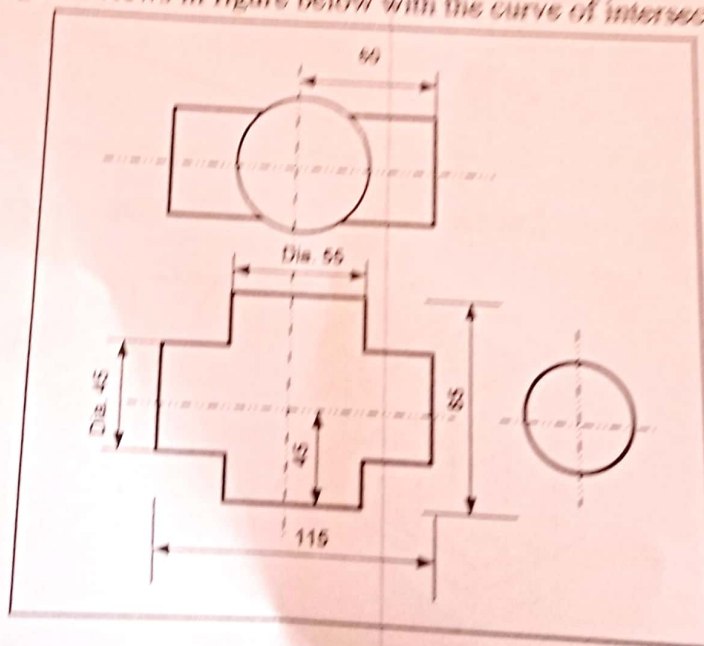


Figure 3

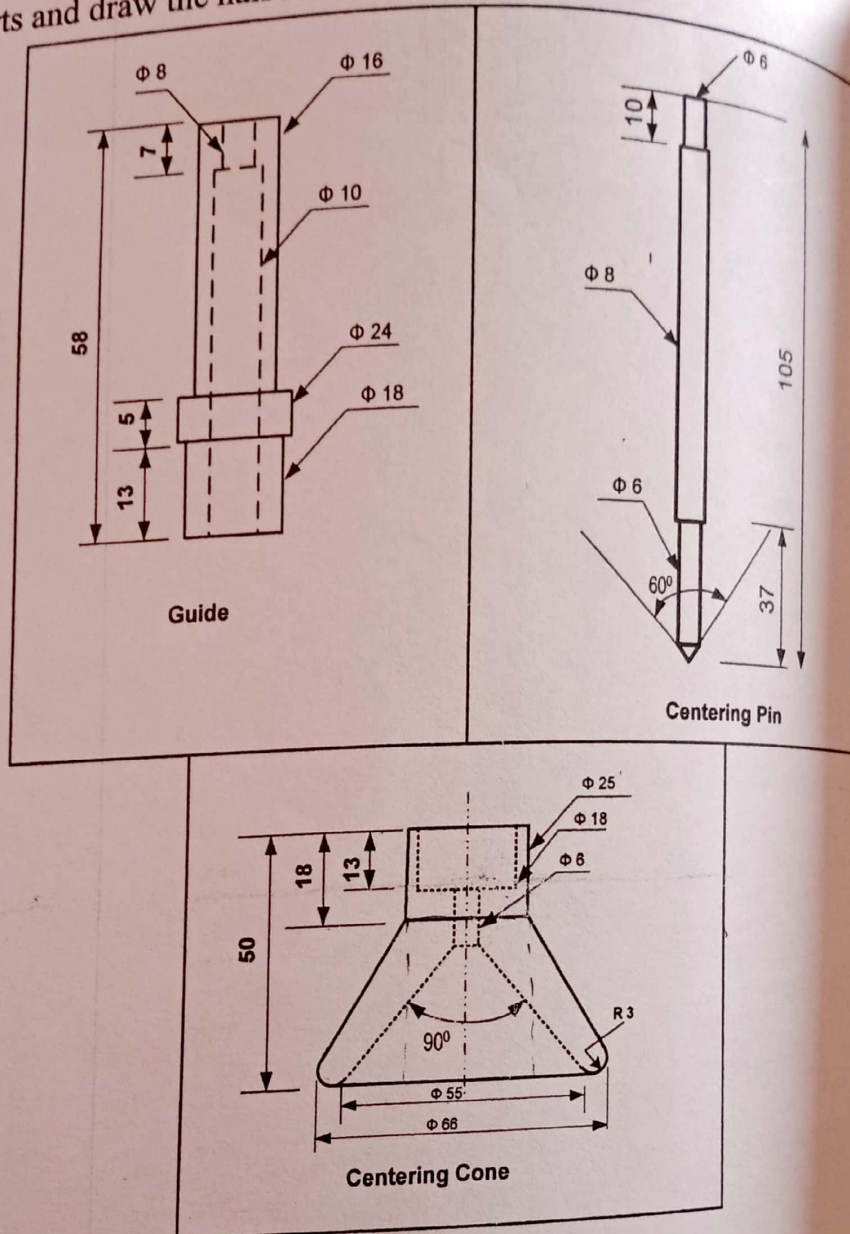
OR

Draw the given views in figure below with the curve of intersection.



5. Figure below shows the detail drawing of a Centering Cone. Assemble the 24

parts and draw the half sectional view.



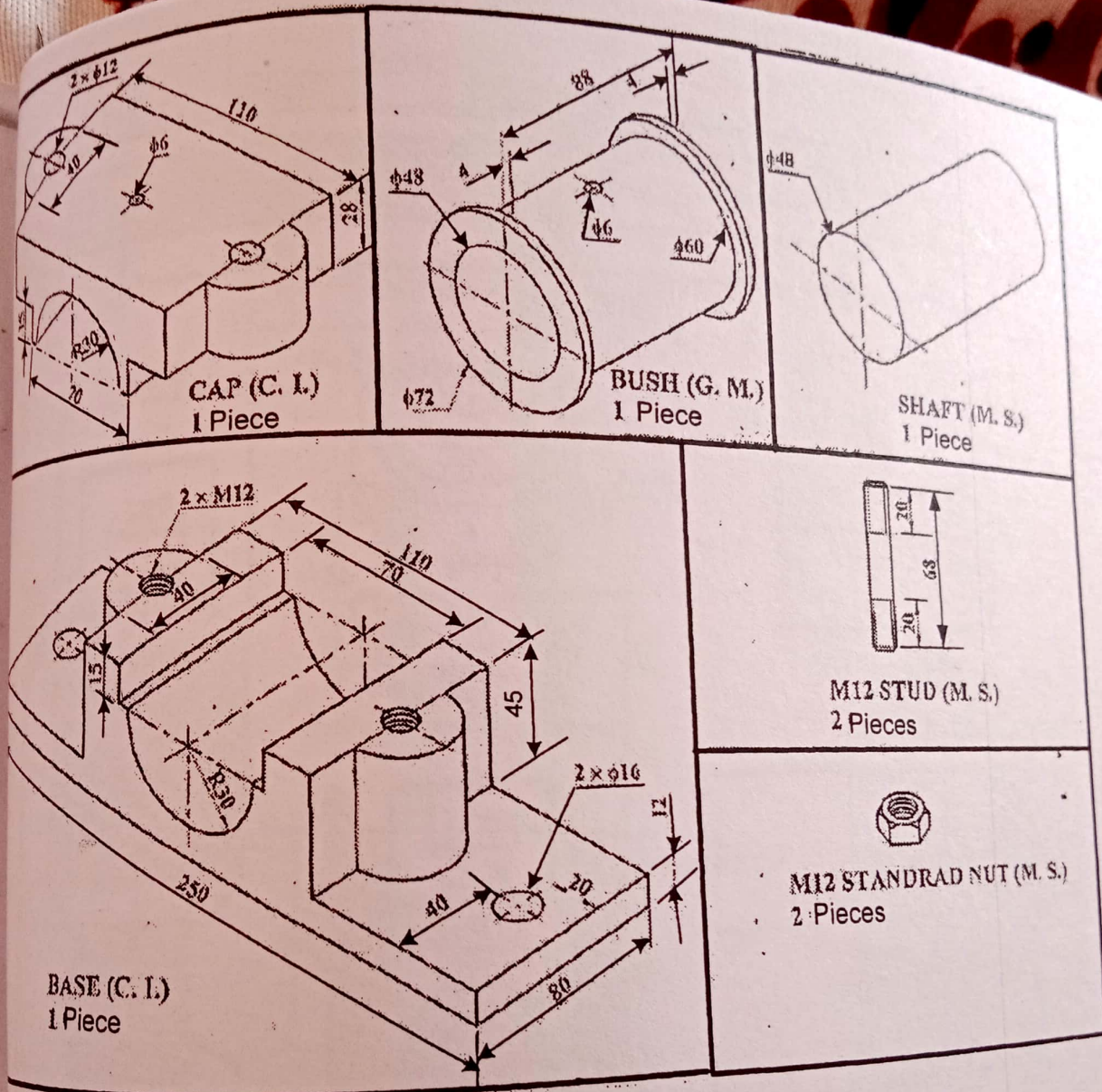


Figure 5

$$\begin{array}{r} 250 \\ 110 \\ \hline 140 \\ 2 \end{array}$$

$$\begin{array}{r} 110 \\ 20 \end{array}$$

Level: Bachelor
Programme: BE
Course: Engineering Drawing

POKHARA UNIVERSITY
Semester: Spring

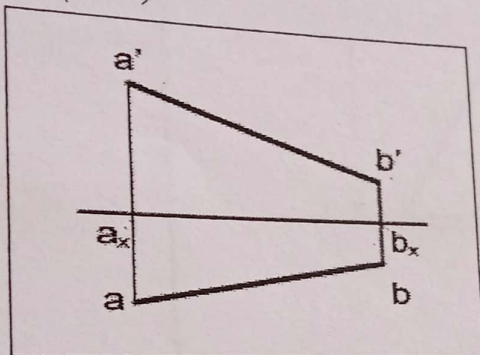
Year : 2013
Full Marks : 100
Pass Marks : 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. Draw an ellipse having major axis 90mm and minor axis 60 mm by definition method. 14
2. Reproduce the given views of the lines and determine the true length. (figure 1) 14



$a'a_x = 35$	$b'b_x = 15$
$aa_x = 20$	$bb_x = 20$
$a_x b_x = 40$	

Figure 1

3. Draw complete orthographic views (three views) of the given object. (figure 2) 30

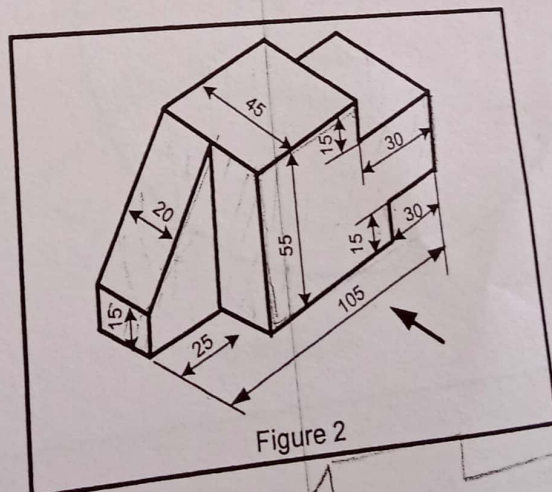
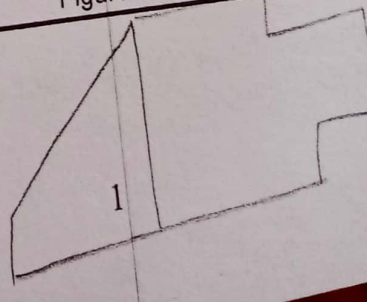


Figure 2



4. A right circular cone is cut by a plane **P** as shown in figure 3. Find the true shape of the section and also construct the lateral surface development of the object. 18

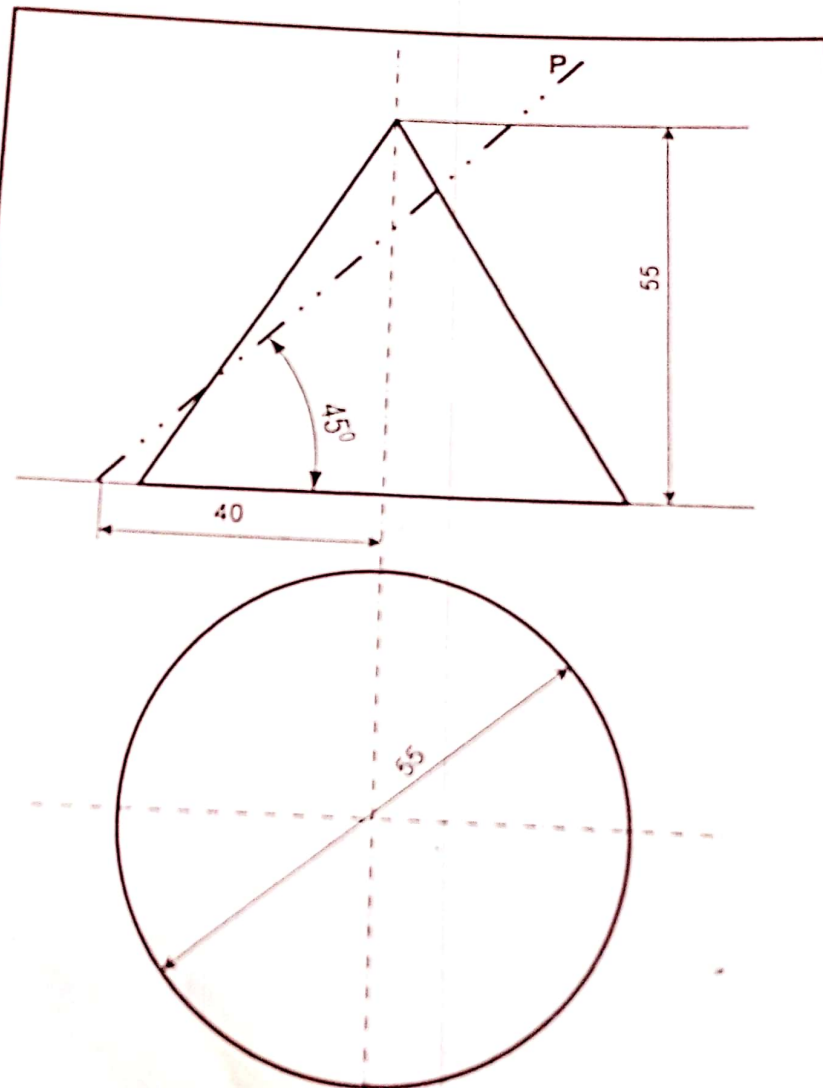
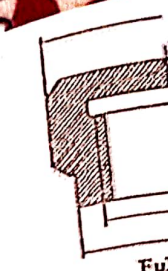


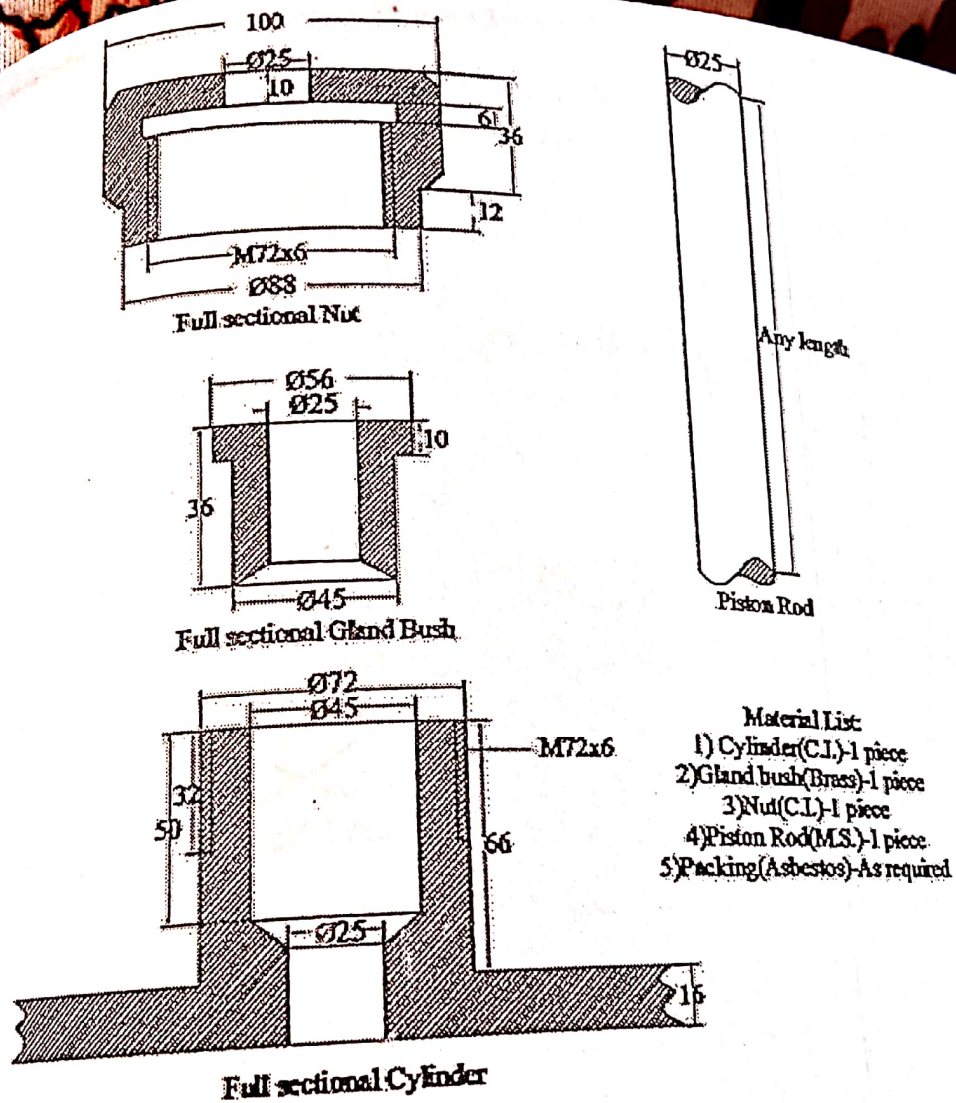
Figure 3

5. Assembled the given parts of the Screw Jack and draw its full sectional (orthographic) front view. (figure 4) 24



OR

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(Figure-5)

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Drawing

Semester: Fall

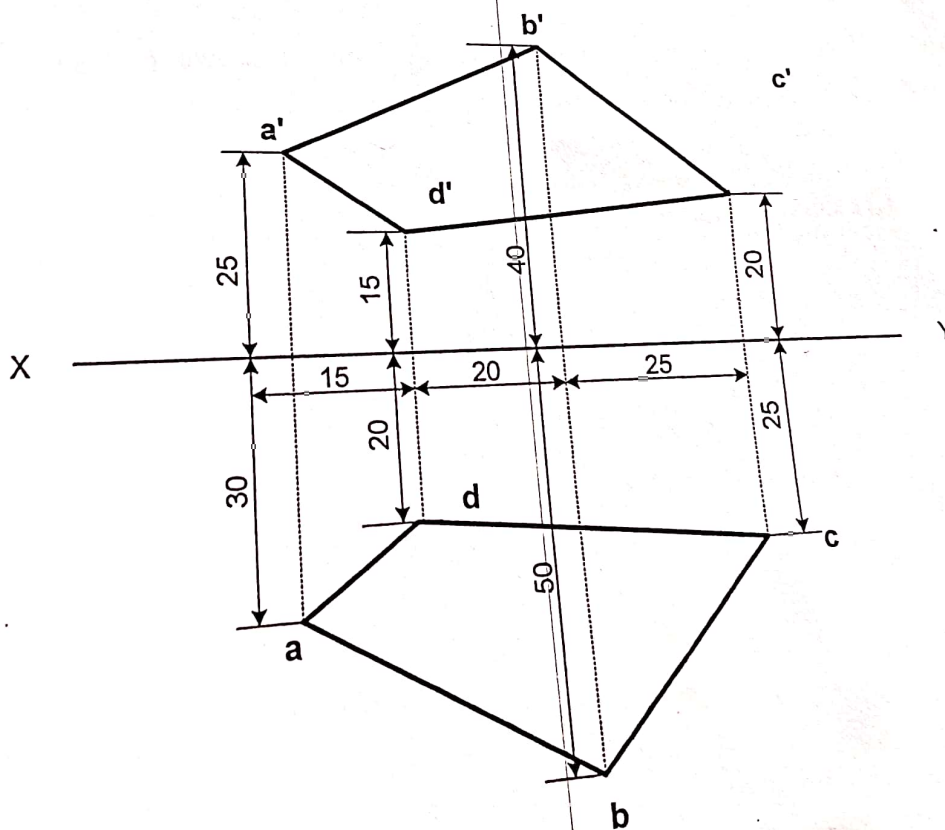
Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

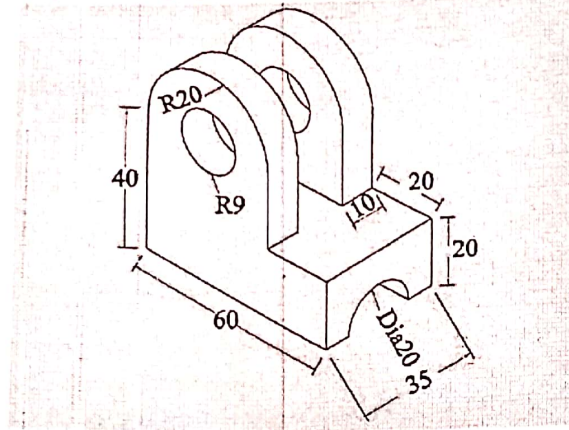
Attempt all the questions.

1. Draw a conical helix of one convolution, where the base diameter of cone is 40 mm and axial height is 60 mm. Also find the top view of the helix. 14
2. Reproduce the given views of plane ABCD and find its true shape. (Figure-1). 14



(Fig.-1)

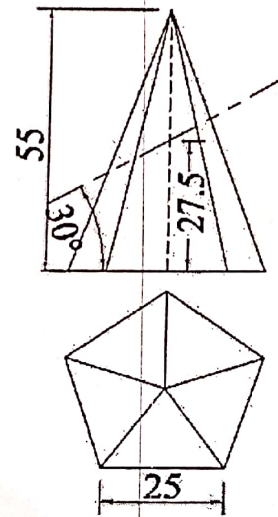
3. Draw the complete orthographic views of the given object. (Figure-2).



(Figure-2)

4. A pentagonal pyramid is cut by a cutting plane as shown in figure-3. Reproduce the given figure and draw:

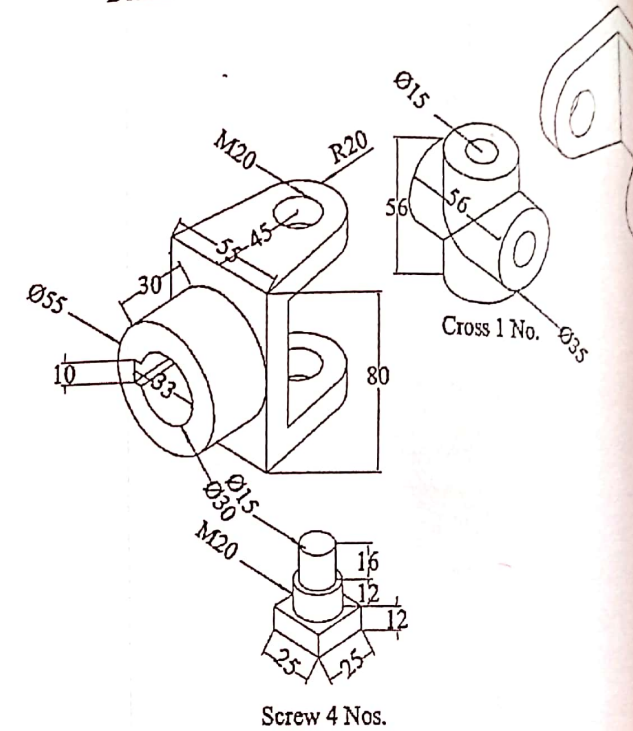
- Sectional top view.
- True shape of the section.
- Lateral surface development of the object.



(Figure-3)

5. Assemble the given parts of Universal Coupling and draw its sectional front view (Figure-4).

Details of Universal Coupling



(Figure-4)

OR

Assemble the given parts of Stuffing Box and draw its full sectional front view (Figure-5).

37

Level: Bachelor
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Course: Engineering Drawing

POKHARA UNIVERSITY

Semester: Spring

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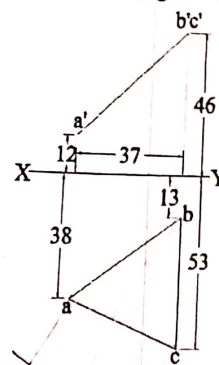
Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

All provided dimensions are in mm.

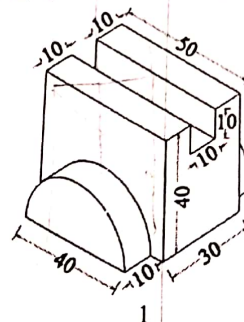
Attempt all the questions.

1. Draw open belt tangent between two circles of 60mm and 30mm diameter, whose centers are 100mm apart. 14
2. Determine the true shape of the plane (figure - 1). 14



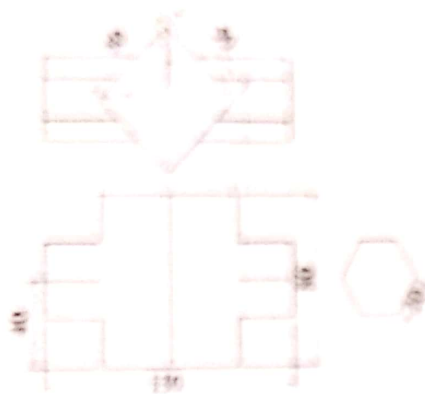
(Figure - 1)

3. Draw the complete orthographic views of the given object (figure - 2). 30



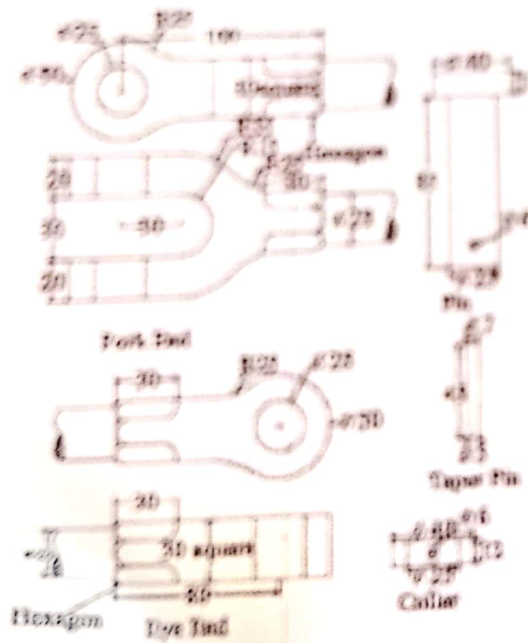
(Figure - 2)

4. Reproduce the given object of the object and draw its full sectional front view (Figure - 3)

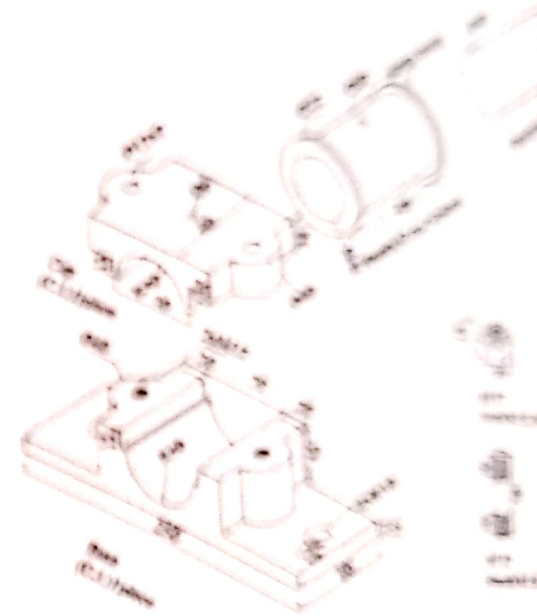


(Figure - 3)

5. Assemble the given parts of kinematic joints and draw its full sectional front view (Figure - 4)

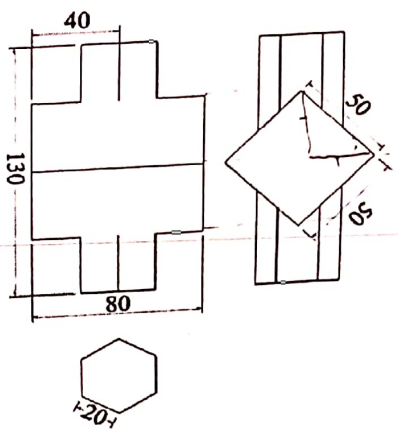


Reproduce the given object of the object and draw its full sectional front view (Figure - 3)



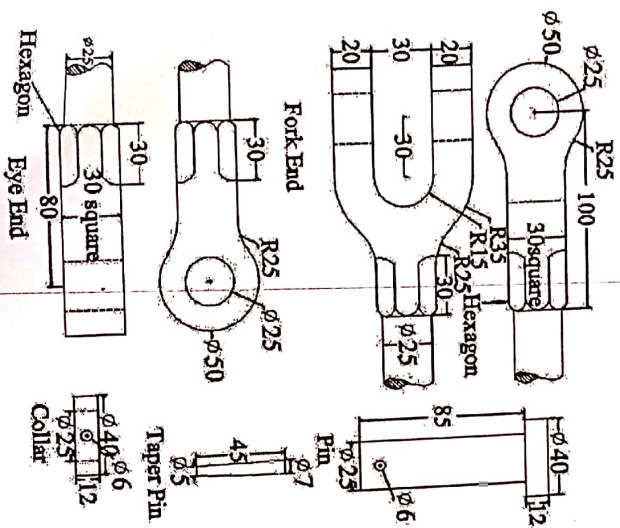
(Figure - 5)

4. Reproduce the given views of the object and draw line of intersection (figure - 3).



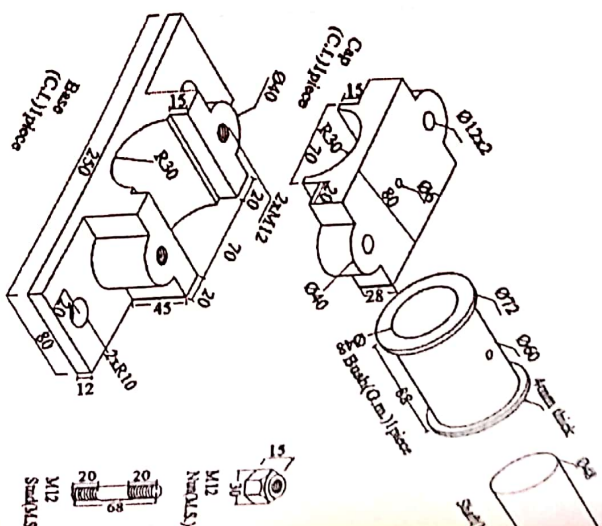
(Figure - 3)

5. Assembled the given parts of Knuckle Joints and draw its full sectional front view (figure - 4).



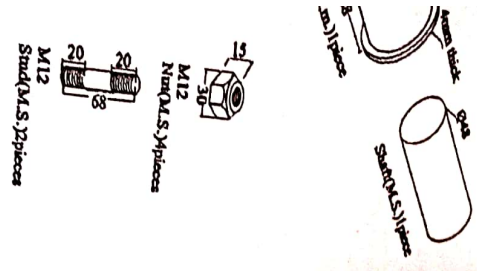
(Figure-4)

Assembled the given parts of Split Bearing and draw its full view (figure-5).

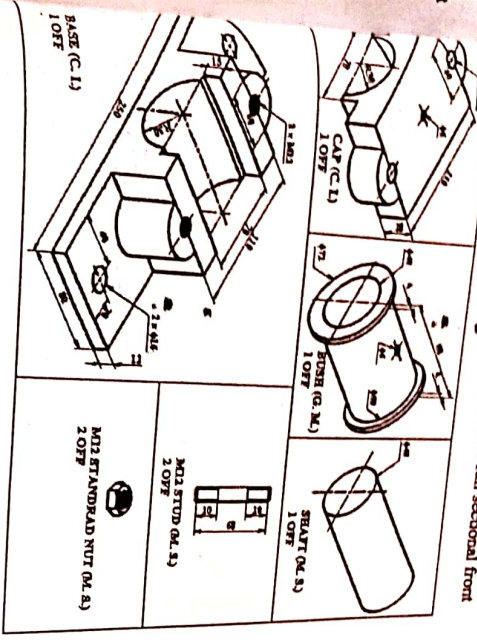


(Figure-5)

(Figure-4)
OR
ring and draw its full sectional front



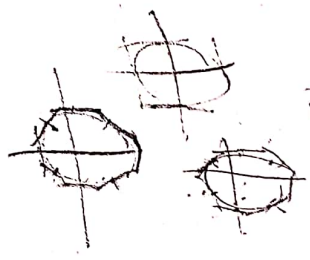
OR
sembled the given parts of split bearing and draw its full sectional front
(Figure-5)



(Figure-5)



4



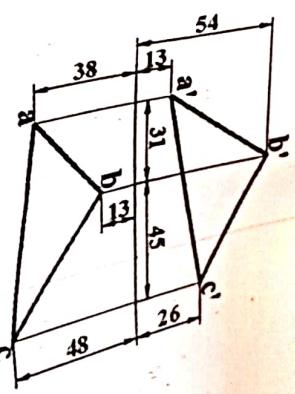
POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Drawing

Semester: Fall
Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

- Construct an uncrossed tangent line between two circles having diameter 50 mm and 30 mm and center to center distance between them is 80 mm.
- Reproduce the given view of the plane and determine the true shape of the plane.
(Figure-1).



(Fig.-1)

- Draw the complete orthographic views of the given object (Figure-2).

30



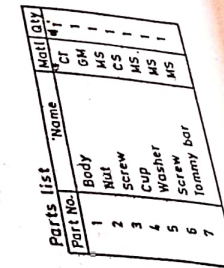
15

- 15



2

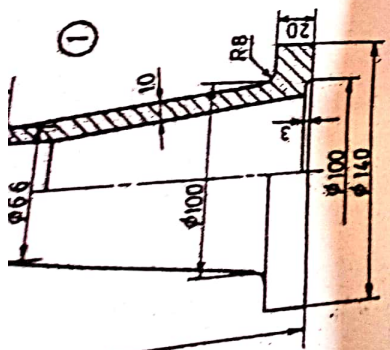
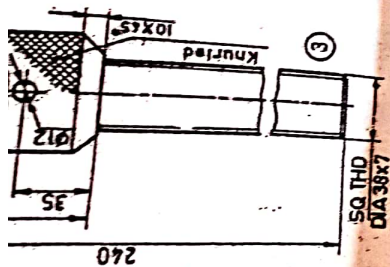
- assemble the given parts of
ew.



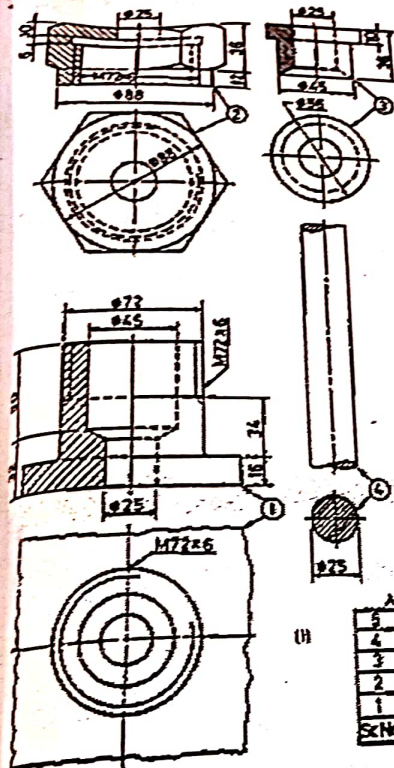
Parts list	Name	Matl	Qty
Part No.		Cr	1
1	Body	GM	1
2	Nut	MS	1
3	Screw	CS	1
4	Cup	MS	1
5	Washer	MS	1
6	Screw bar	MS	1
7	Tommy bar		



	Name	Material
1	Body	Cr
2	Nut	GM
3	Screw	MS
4	Cup	CS
5	Washer	MS
6	Screw	MS
7	Tommy bar	MS



Use the given parts of Stuffing Box and draw its half sectional front



SCALE: 1:0.8

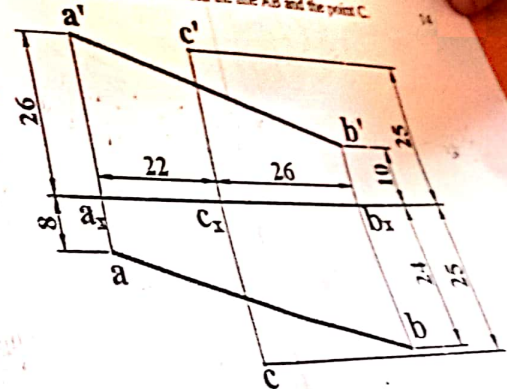
ALL DIMENSIONS IN MM				
5	Packling		Asbestos	
4	Piston rod	1	C-30	
3	Gland bush	1	Brass	
2	Nut	1	CL	
1	Cylinder	1	CL	
S.No	Name of part	No. of	Material	Remarks

4

Year : 2004
Full Marks: 100
Pass Marks: 45
Time : 3hrs

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

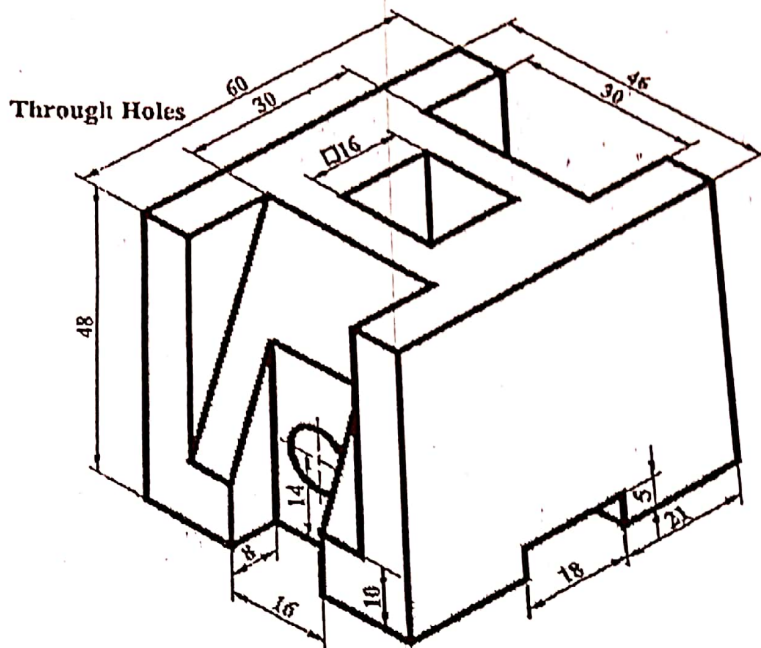
- Construct one turn of cylindrical helix with diameter of 40 mm and pitch of 60 mm.
- Determine the shortest distance between the line AB and the point C.



3. Draw complete orthographic views of the given object.

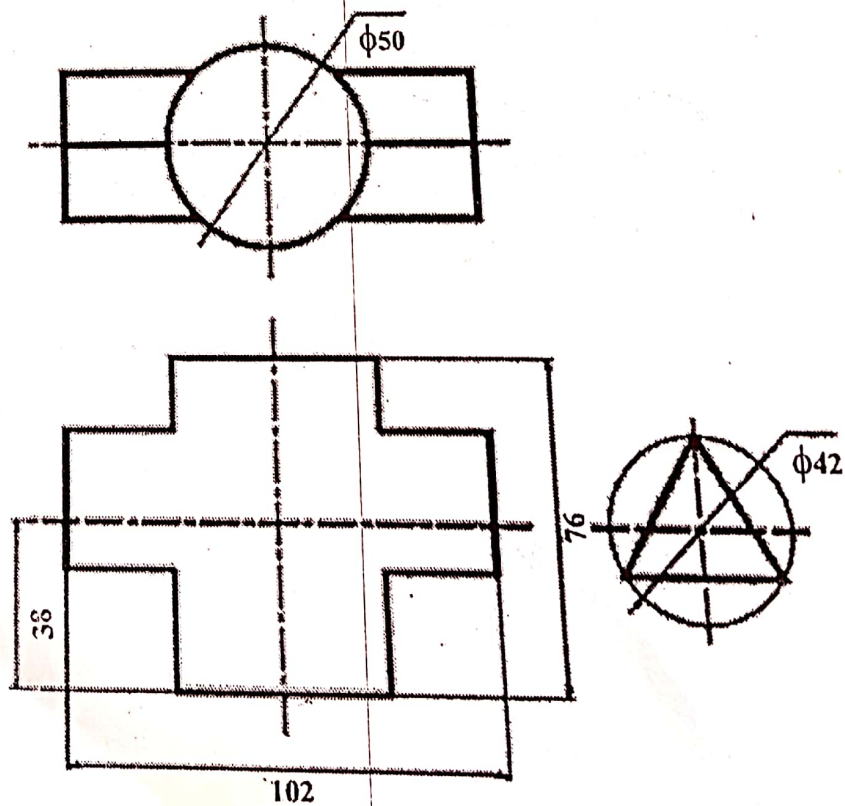
30

5. The
con
view

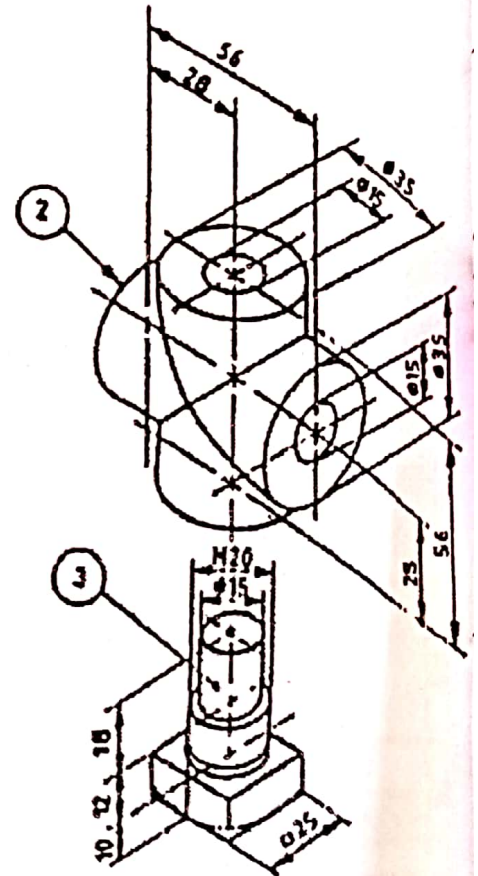
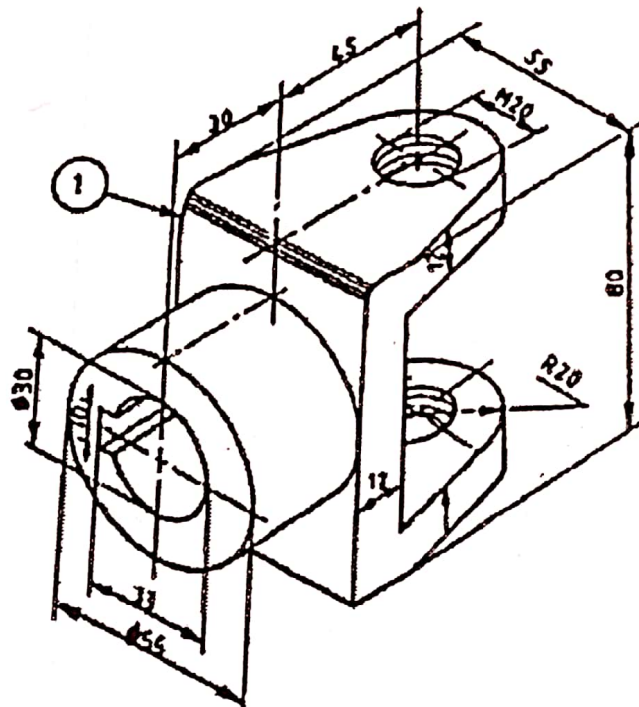


4. Draw the given views in the figure below with curve of intersection.

18



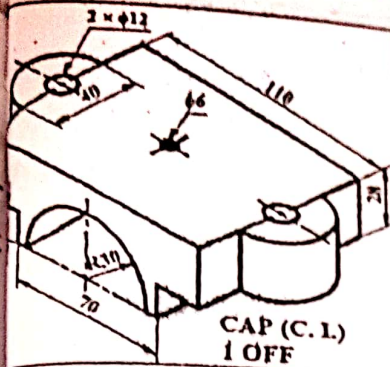
5. The figure below shows detail drawing of a UNIVERSAL COUPLING for connecting two shafts. Assemble the parts and draw the sectional front view of the assembly.



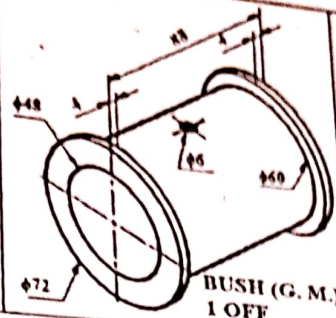
PARTS LIST

No	Name of part	Qaunt
1	Fork	2
2	Cross	1
3	Screws	4

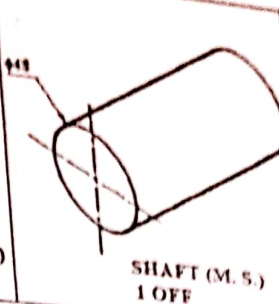
UPLING for
ctional front



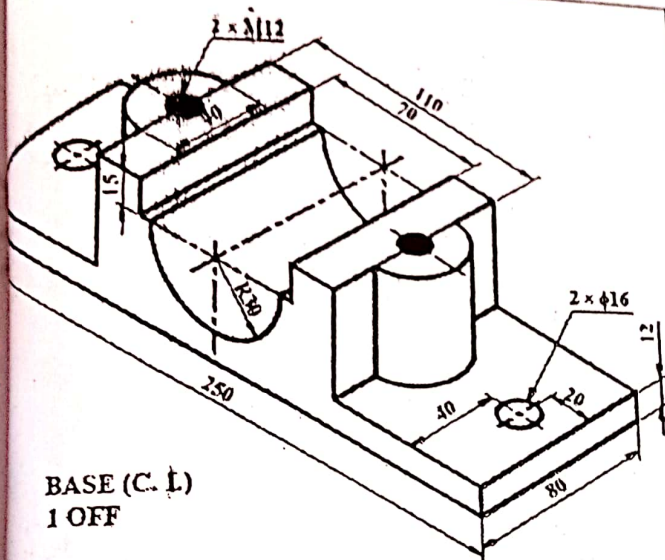
CAP (C. I.)
1 OFF



BUSH (G. M.)
1 OFF



SHAFT (M. S.)
1 OFF



BASE (C. I.)
1 OFF



M12 STUD (M. S.)
2 OFF



M12 STANDRAD NUT (M. S.)
2 OFF

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Drawing

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

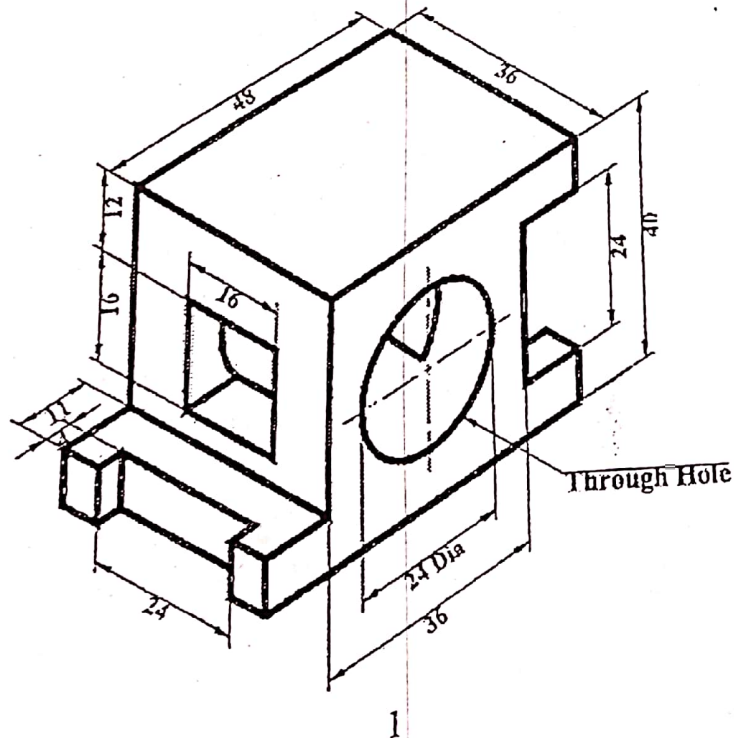
Attempt all the questions.

1. Construct a crossed line tangent (crossed belt type) between two circles having radii 20 mm and 15 mm and centre to centre distance between them is 75 mm. 14
2. a) Draw true shape of given triangular plane from given co-ordinates. 8

	X	Y	Z
A	75	20	10
B	40	10	45
C	0	50	20

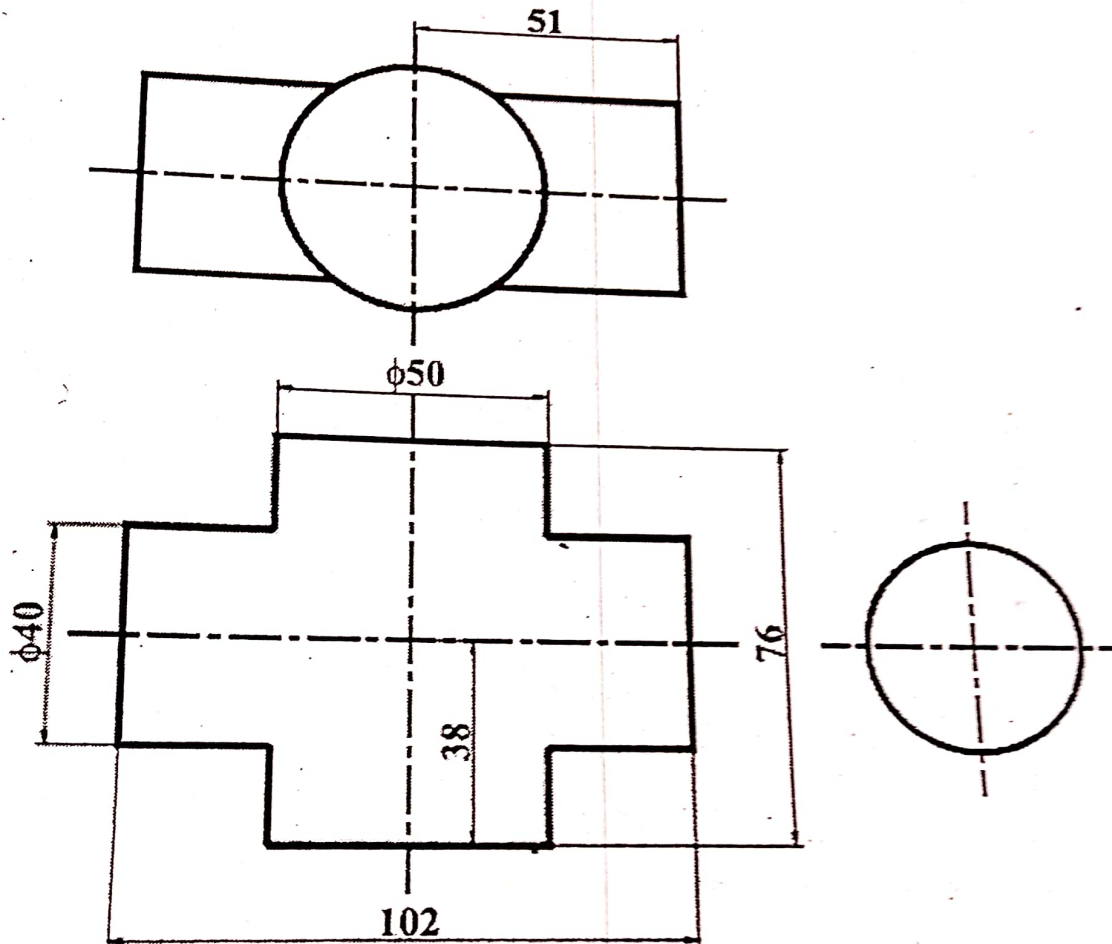
MM

- b) Draw an octagon having side of 30 mm. 6
3. Draw complete orthographic views of the given object. 30

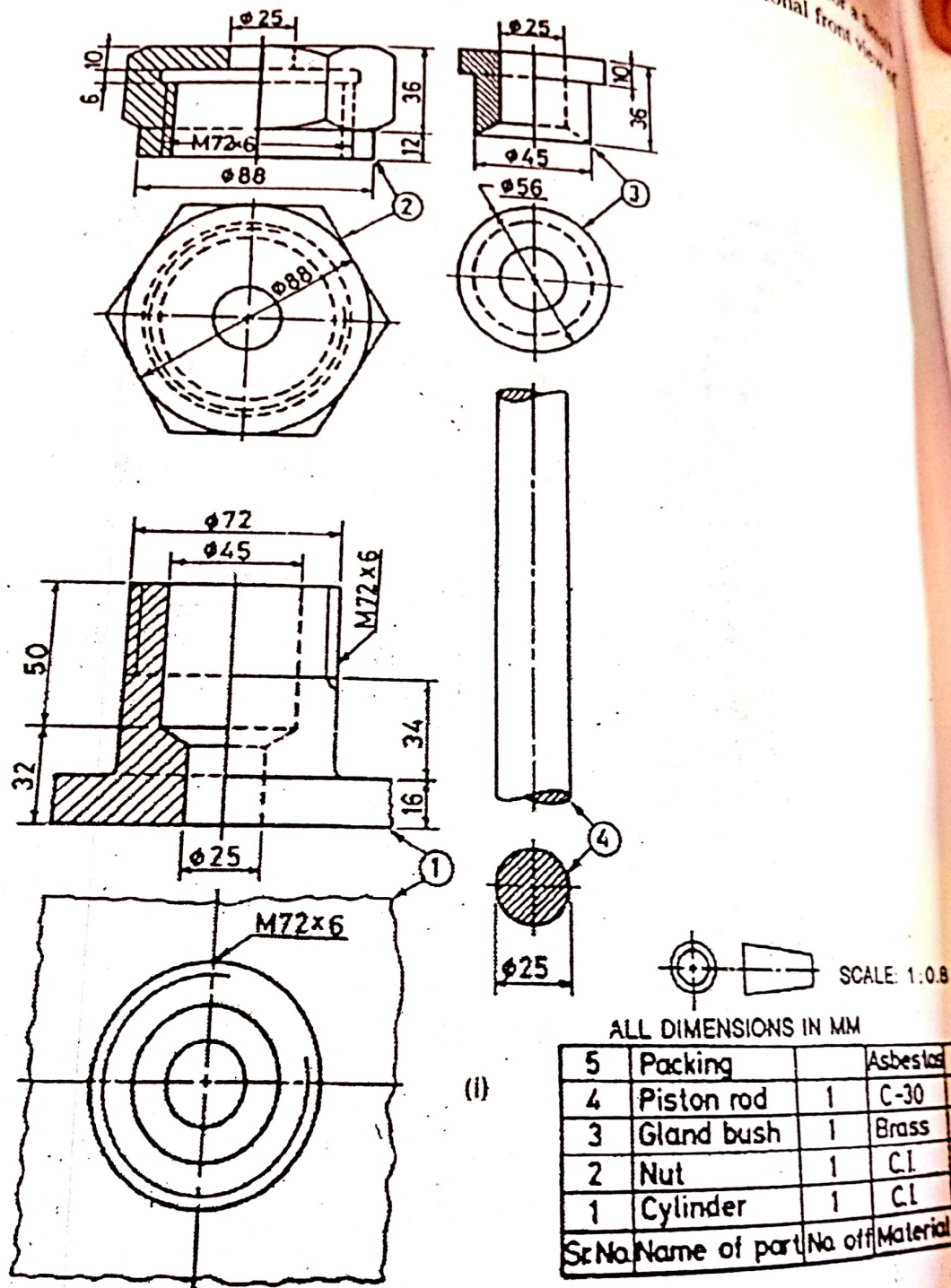


Draw the given views of the figure below with curve of intersection.

12



5. The figure below shows detail drawing of a STUFFING BOX for a Steam Engine. Assemble the parts and draw the half sectional front view of the assembly.



OR

The detail drawing of a SPLIT BEARING is shown in the figures below. Draw the assembled front view with section. Take any length for the shaft.

IN MM	ASBESTOS
1	C-30
1	Brass
1	CI
1	CI
1	CI
No off	Material Remo

SCALE: 1:0.8

ING BOX for a Small
sectional front view o

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Engineering Drawing

Semester: Spring

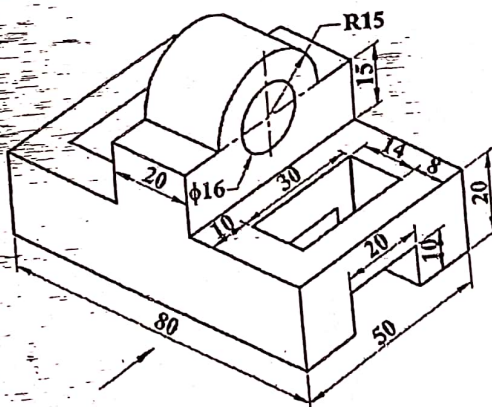
Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to make neat and accurate drawings with dimensions.

The figures in the margin indicate full marks.

Attempt all the questions.

- Construct a conic when the distance of its focus from its directrix is equal to 50 mm and its eccentricity is $\frac{3}{4}$. Measure its major and minor axes. Draw a tangent at any point on the curve. What is the distance between the foci? 14
- Draw a line 50 mm long and trisect it. 6
- A line 40 mm long is parallel to V.P and inclined at an angle of 30° to H.P. The end A is 15 mm above H.P and 20 mm in front of V.P. Draw the projections of the line. 8
- Draw complete and neat orthographic views of the following figure. 30



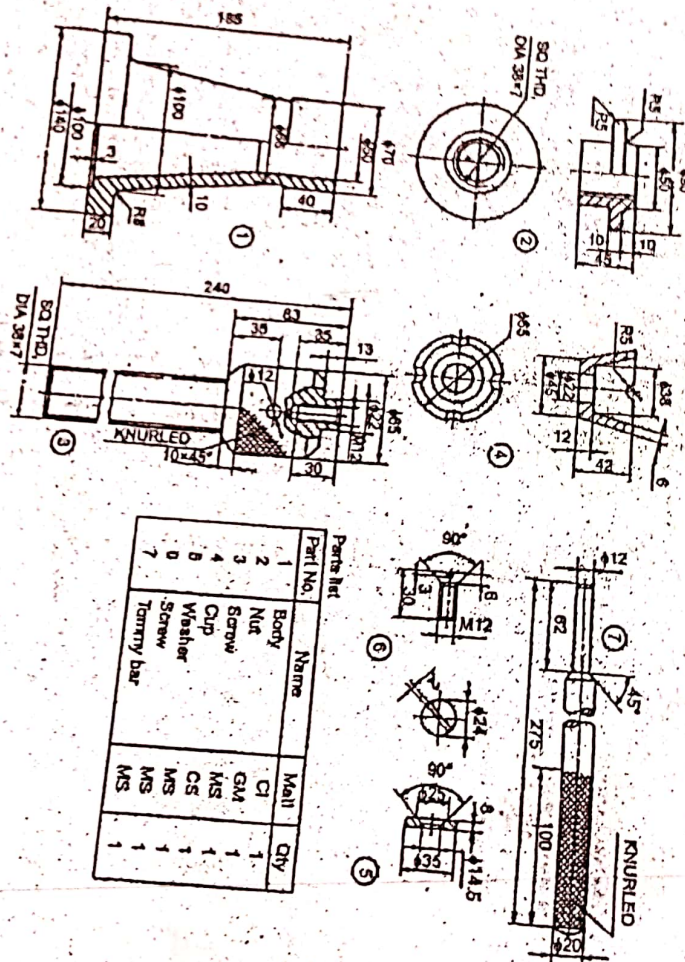
- A hexagonal pyramid with side of base 30 mm and height 75 mm stands with its base on H.P and an edge of the base parallel to V.P. It is cut by a plane perpendicular to V.P, inclined at 45° to H.P and passing through the 18

mid-point of the axis. Draw the (sectioned) top view and develop the lateral surface of the truncated pyramid.

OR

A vertical cylinder of diameter 120 mm is fully penetrated by a cylinder of diameter 90 mm, their axes intersecting each other. The axis of the penetrating cylinder is inclined at 30° to the H.P and is parallel to the V.P. Draw the top and front views of the cylinders and the curves of intersection.

6. Assemble all the parts of screw Jack and draw the half sectional view from the front.



Level: Bachelor
 Programme: BE
 Course: Engineering Drawing

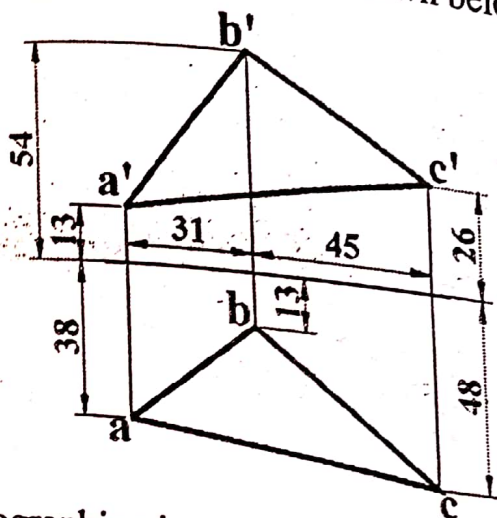
Semester: Fall

Year : 2017
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

4. Draw the intersection

Candidates are required to draw neat and clear figures with appropriate dimensions.
 The figures in the margin indicate full marks.
 Attempt all the questions.

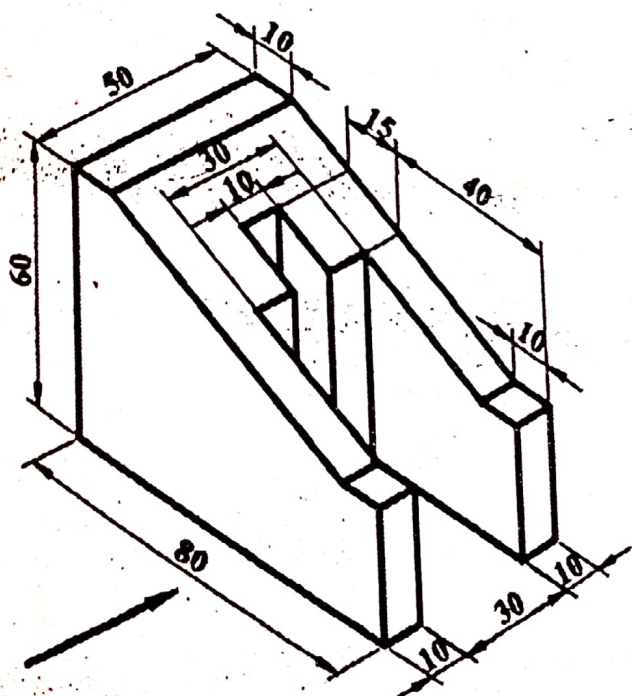
Draw one turn of a helix of pitch 60 mm on a cylinder of diameter 40 mm.
 Draw the true shape of a plane for a figure shown below.



14

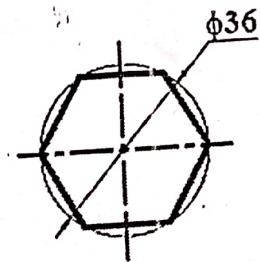
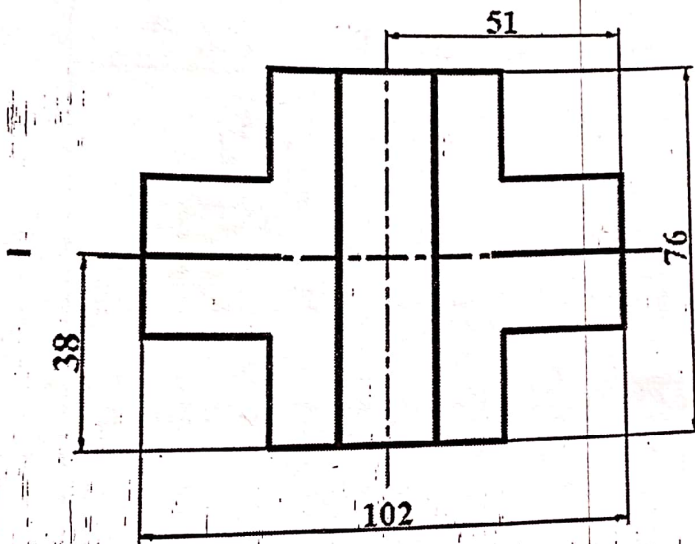
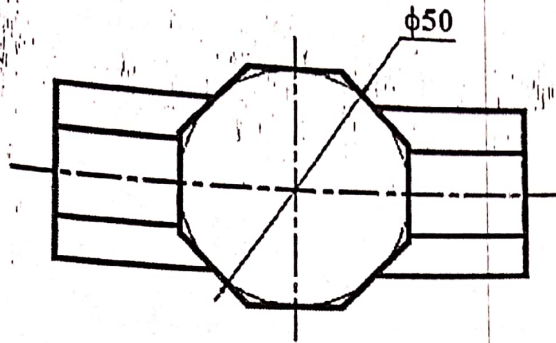
14

Draw complete orthographic views of the given object.

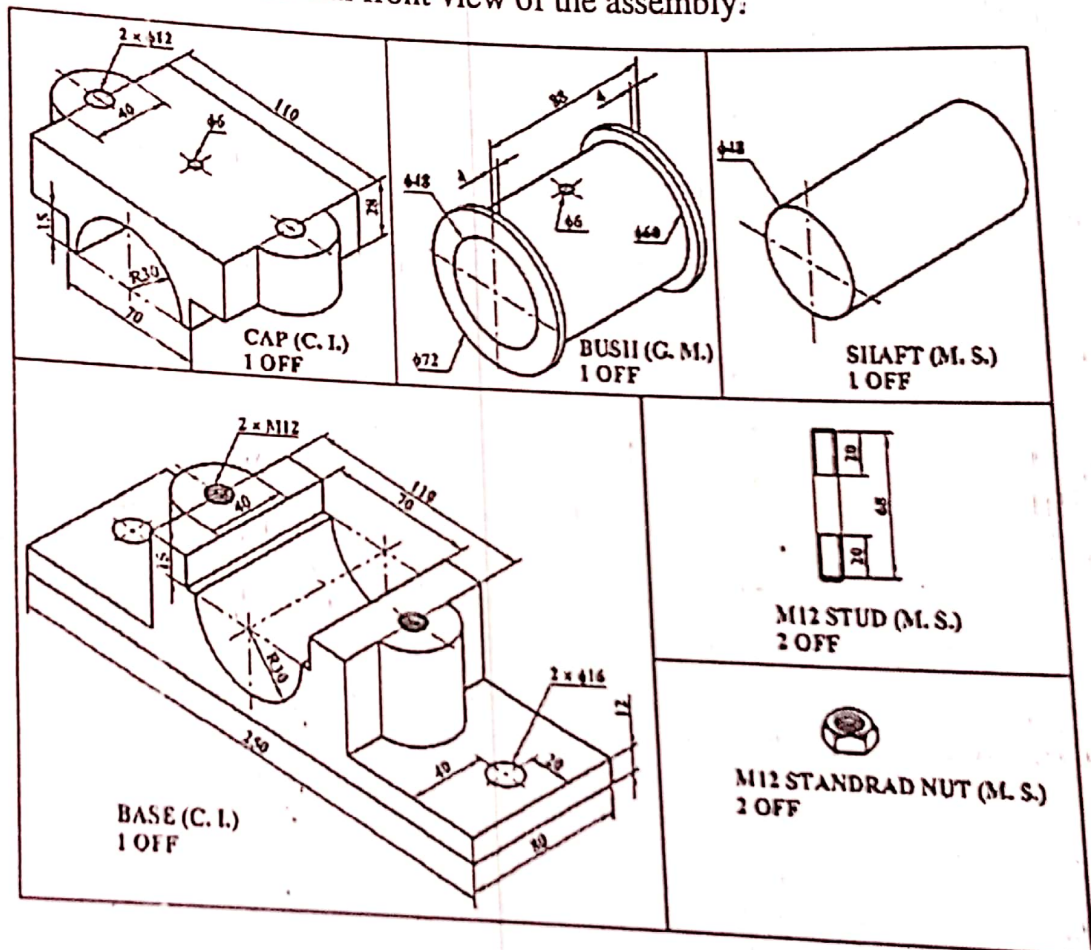


30

4. Draw the complete orthographic drawing showing the curves of intersection of solid for the figure shown below. 18



5. The figure below shows the detail drawing of the parts. Assemble the parts and draw the half sectional front view of the assembly: 24



POKHARA UNIVERSITY

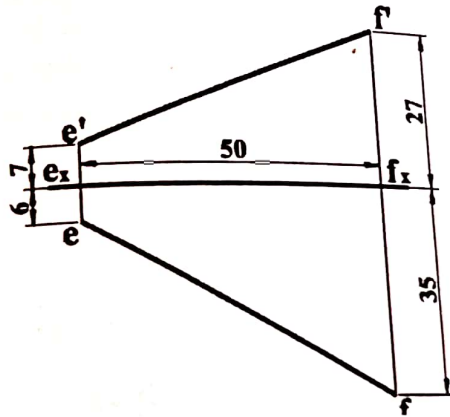
Semester: Fall

Level: Bachelor
Programme: BE
Course: Engineering Drawing

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

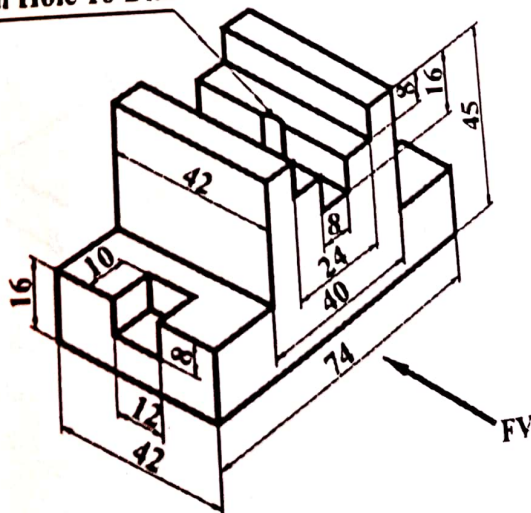
Candidates are required to draw neat and clear figures with appropriate dimensions.
The figures in the margin indicate full marks.
Attempt all the questions.

1. Draw indirect common tangent to two circles of diameter 40 mm and 60 mm with centre to centre distance of 120 mm. 14
2. Produce the given views of the lines and determine the true length. 14



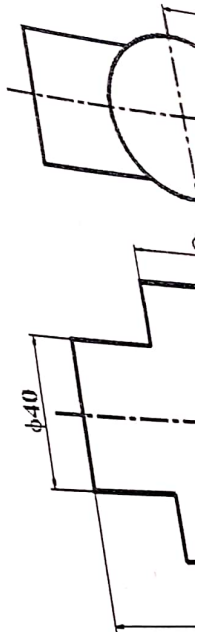
3. Draw complete orthographic views of the given object. 30

Through Hole 10 Dia.

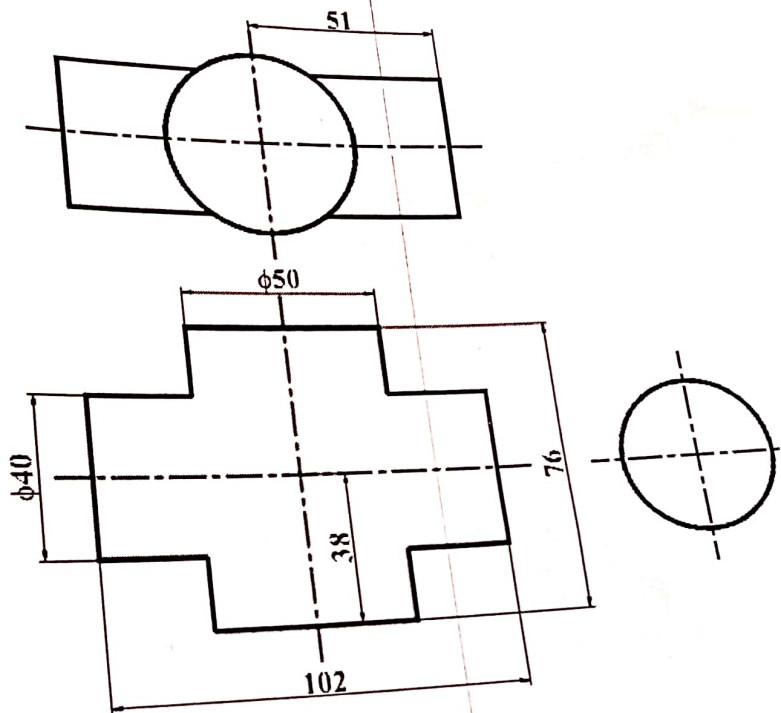


1

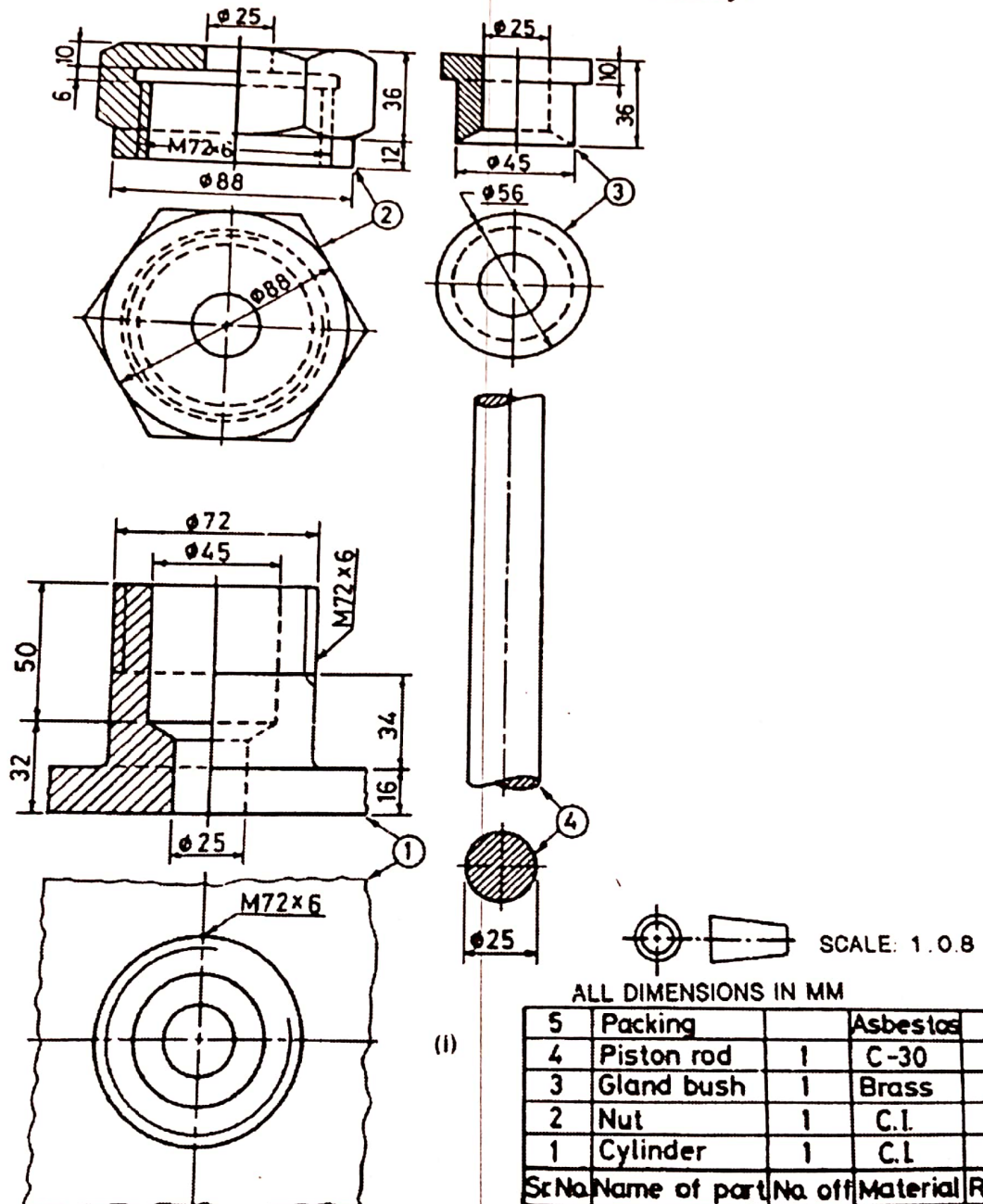
4. Draw the complete orthographic intersection of solid for the figure shown.



4. Draw the complete orthographic drawing showing the curves of intersection of solid for the figure shown below.



5. The figure below shows the detail drawing of the parts. Assemble the parts and draw the half sectional front view of the assembly. 24



POKHARA UNIVERSITY

Semester: Fall

Level: Bachelor
Programme: BE
Course: Engineering Drawing

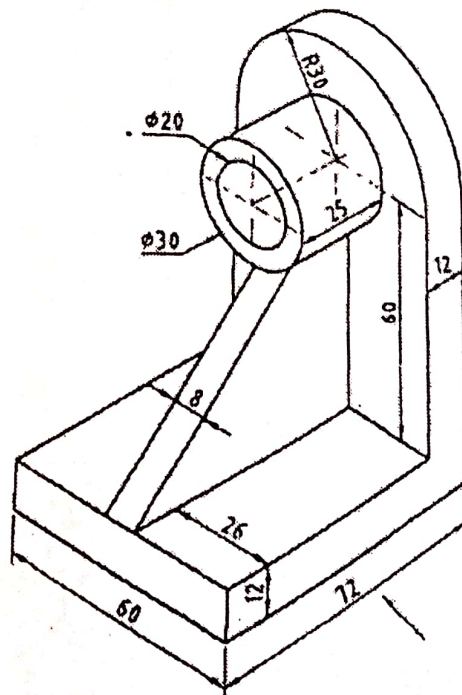
Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to make neat drawings with accurate dimensions.
The figures in the margin indicate full marks.
Assume suitable data if necessary.
Attempt all the questions.

4 Draw the given views of intersection.

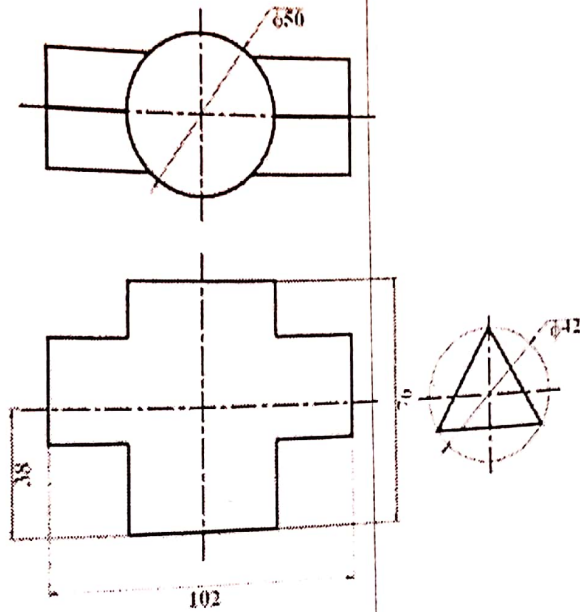


- 14 Draw a hypocycloid having a generating circle of diameter 50 mm and directing circle of radius 10 mm. Also draw a normal and a tangent at any point M on the curve.
- 14 A line AB, 90 mm long, is inclined at 30° to the H.P. Its end A is 12 mm above the H.P. and 20 mm in front of the V.P. Its front view measures 65 mm. Draw the top view of AB and determine its inclination with the H.P.
- 30 Draw complete and neat orthographic views of the following figure.



5 The figure below shows parts and draw the half-section.

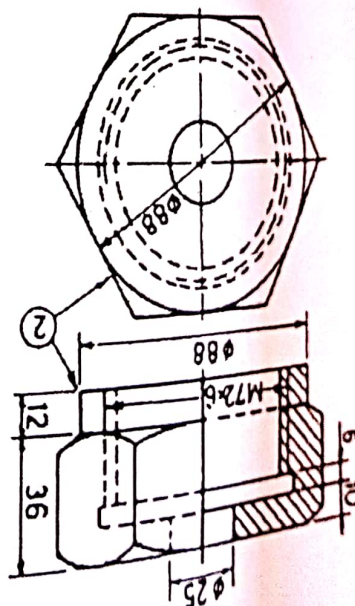
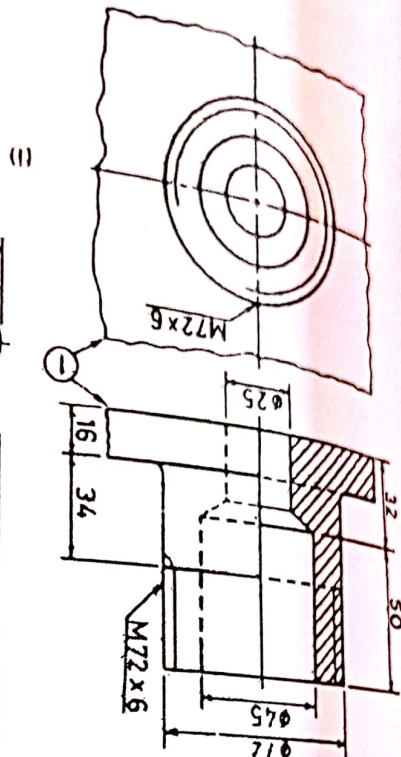
- 4 Draw the given views of assigned form and complete the intersection. 18



- 5 The figure below shows the detail drawing of the parts. Assemble the parts and draw the half sectional front view of the assembly. 24

ALL DIMENSIONS IN MM

SCALE 1:0.8



Candidates are required to use appropriate dimensions for the figures in the main text. Assume suitable data where necessary. Attempt all the questions.

Attempt
Draw a cycloid, given
Also draw a tangent
Distance between
End A is 40 mm
behind the VP. 7
projections. Also c
Draw the comple
(figure 1).

Candidates are required to draw neat and accurate drawings in 3 hrs.
 appropriate dimensions.
 the figures in the margin indicate full marks.
 assume suitable data if necessary. All dimensions are in mm.
 attempt all the questions.

- Draw a cycloid, given the diameter of a generating circle as 50 mm. 14
 Also draw a tangent and normal at any given point T on the curve.
 Distance between the end projectors of a straight line AB is 40 mm. 14
 End A is 40 mm behind VP and 20 mm below HP and end B is 6 mm
 behind the VP. The line is inclined at 30° to the VP. Draw its
 projections. Also obtain its TL and its inclination (θ).
 Draw the complete orthographic views (3 views) of the given object 30
 (figure 1).

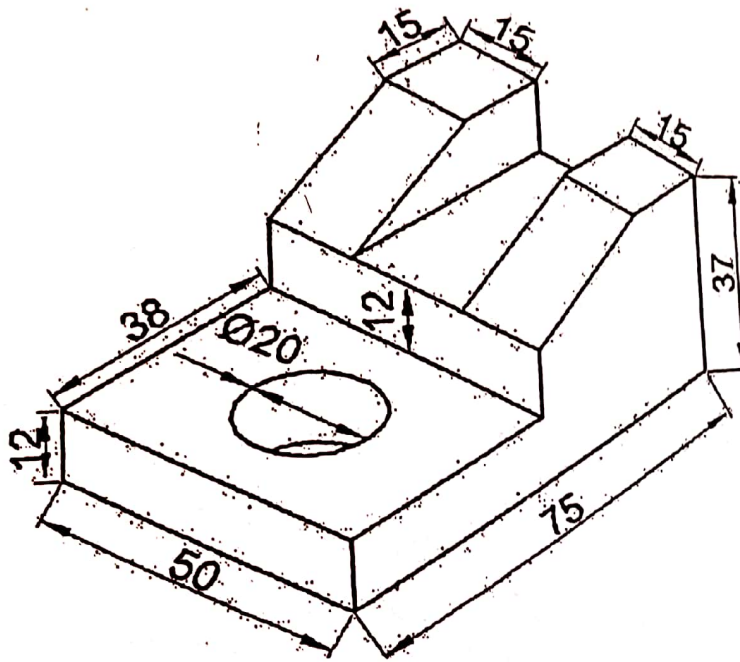


Figure 1

4. Reproduce the given forms of the object and draw curve of intersection (figure 2).

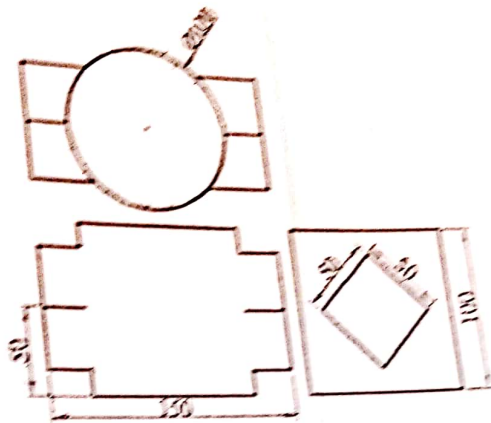
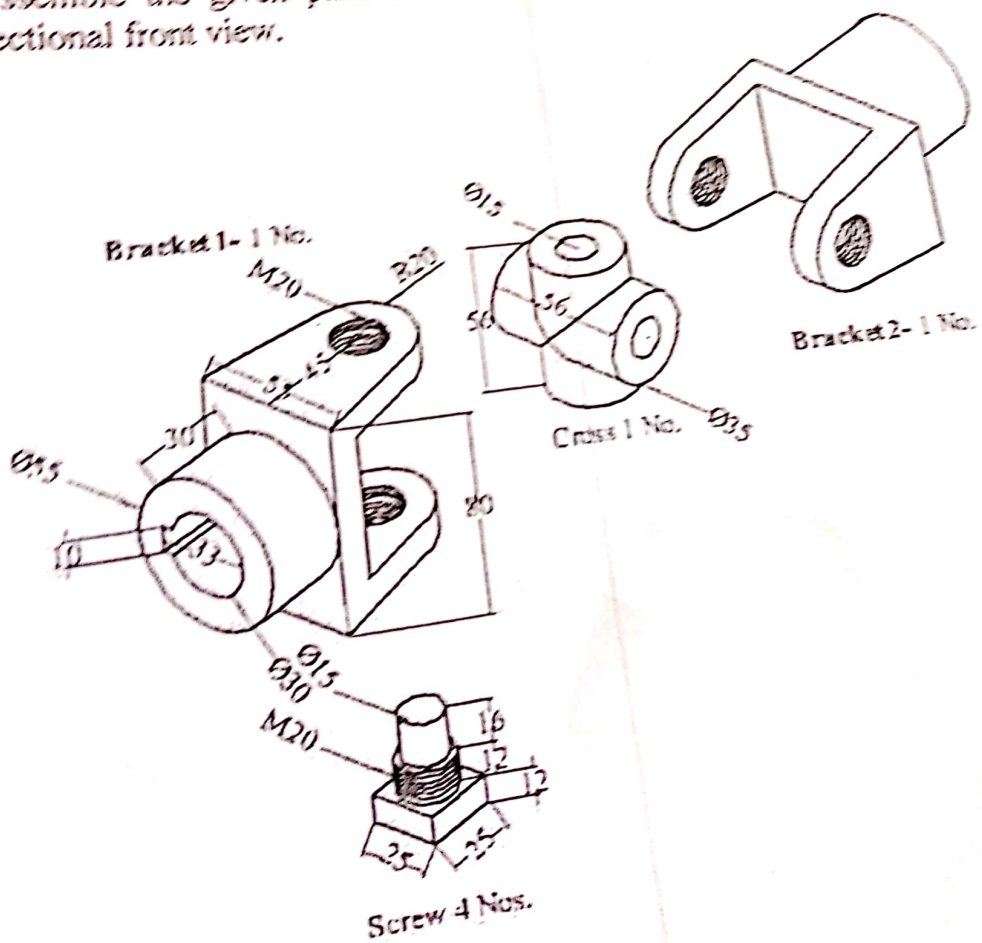


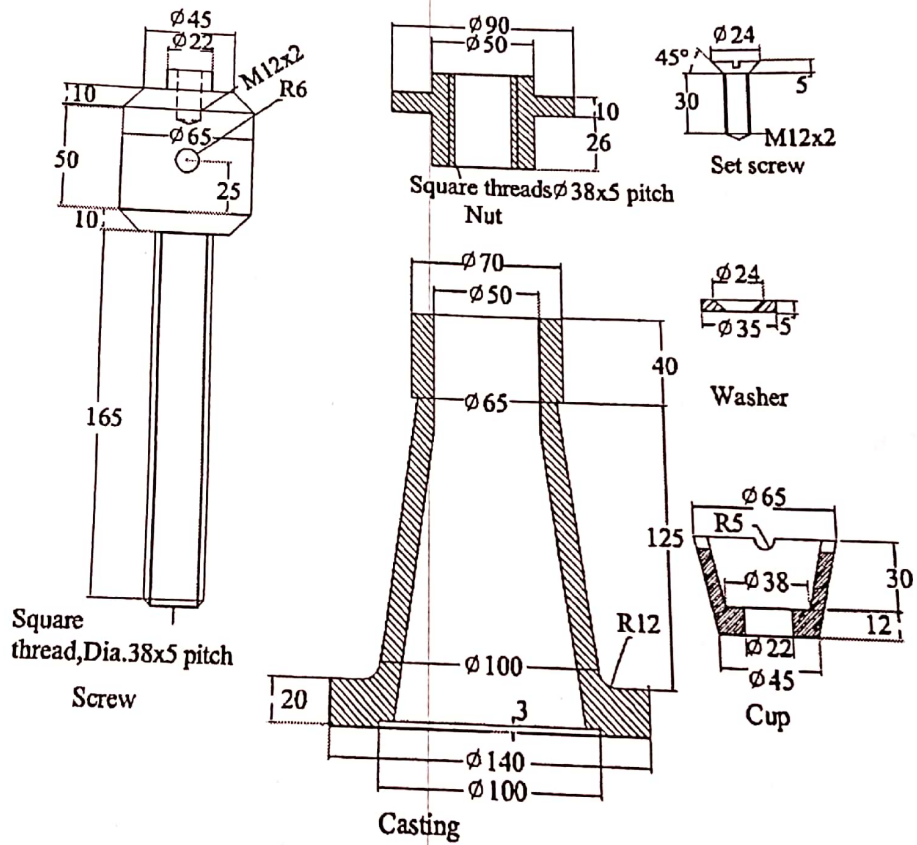
Figure 2

5. Assemble the given parts of Universal Coupling and draw its full sectional front view. 24



OR

Assemble the given parts of Screw Jack and draw its full sectional front view.





Year : 2019

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

Answers are required to give their answers in their own words as far as possible.

Marking scheme is applicable.

Figures in the margin indicate full marks.

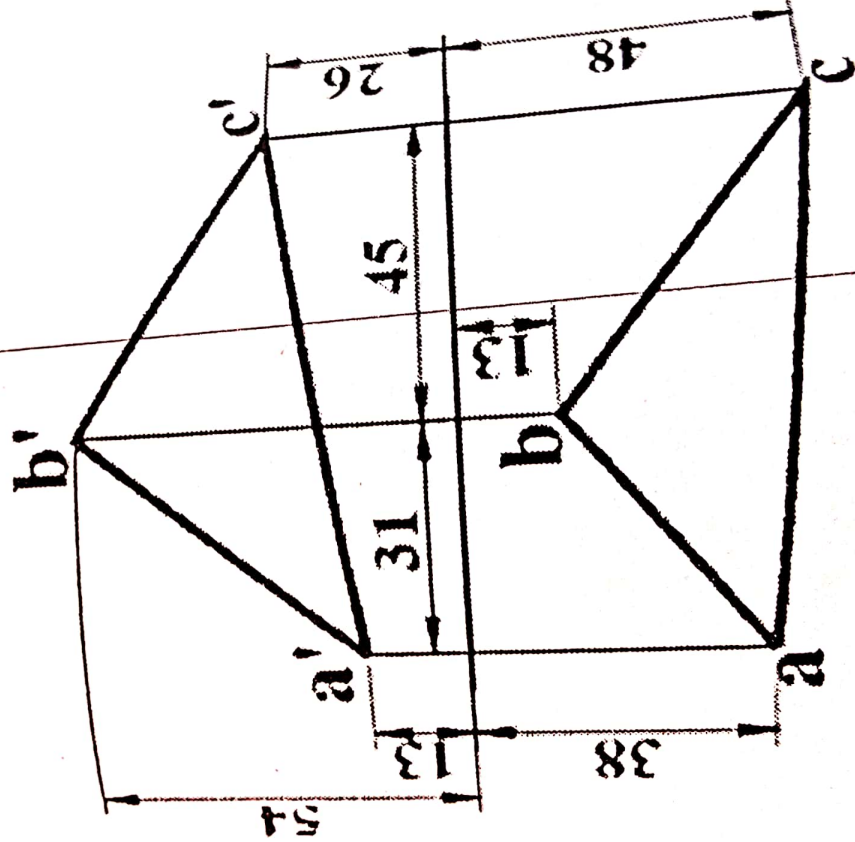
Attempt all the questions.

1. A hypocycloid, given the radii of rolling and directing circles as $r=20$ and $R=80$ mm, respectively. Also draw a normal and a tangent at any point on the curve.

14

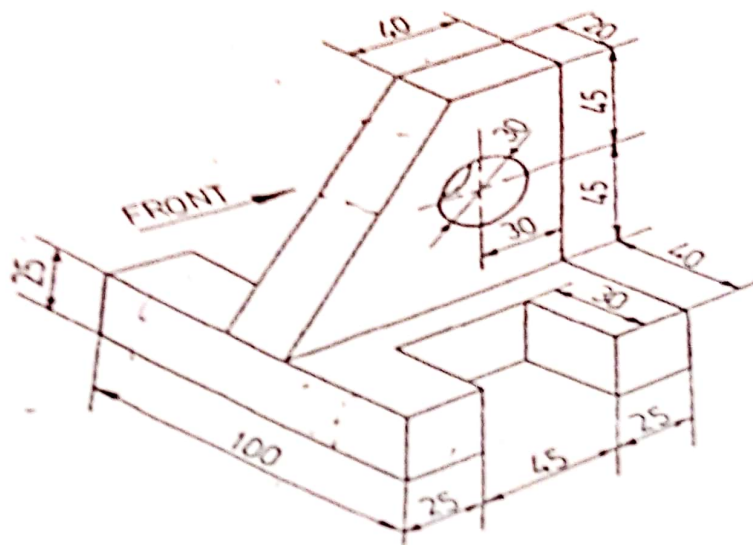
2. Produce the given view of the plane and determine the true shape of the

14



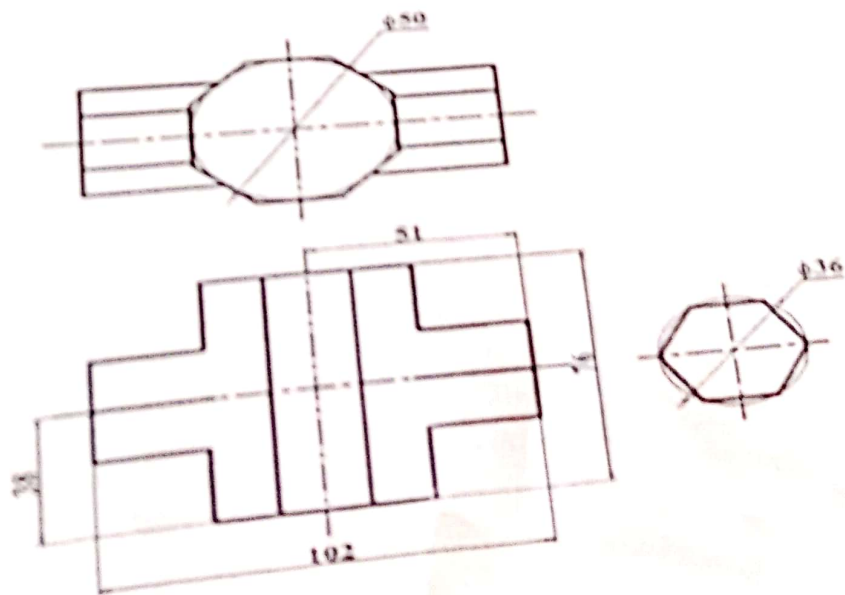
3. Draw the complete orthographic views of the given object.
(Assume suitable data if necessary)

30



4. Draw the given views of the figure below with the curves of intersection.

18



- 24



OR

Technical drawing of a mechanical assembly showing four components:

- Bracket 1- 1 No.**: A 3D view of a bracket with dimensions: $\phi 55$, 30, 10, 45, 80, $M20$, $R20$, and 56.
- Bracket 2- 1 No.**: A 3D view of a bracket with dimensions: $\phi 15$, 56, and $\phi 35$.
- Cross 1 No.**: A 3D view of a cross with dimensions: $\phi 15$, 56, and $\phi 35$.
- Screw 4 Nos.**: A 3D view of a screw with dimensions: 16, 12, 12, 25, 25, and $M20$.